

Dynamic Outside Options and Optimal Negotiation Strategies[†]

By ANDREW MCCLELLAN*

We study the design of negotiation strategies when a principal and agent must decide how to split a pie while the agent's outside option changes over time. The principal's optimal strategy under commitment demonstrates a new, but intuitive, set of negotiation dynamics. When the agent is tempted to leave, the principal gradually promises a larger share (decreasing demands) and more time to explore the outside option (decreasing pressure), illustrating a complementarity between these two tools. Although the principal's expected utility is decreasing in the outside option, his expected utility and demands are increasing in the outside option's drift and volatility. (JEL D63, D82, D86)

Changing outside options are an important factor in dynamic negotiations between organizations and agents. Workers can receive competing offers while negotiating a contract with a firm; the market value of a company may change while it is in the process of being acquired in a merger, changing the value to the acquirer of pulling out of the merger; during sovereign debt negotiations, the value to a country of leaving negotiations depends on economic fundamentals that evolve over time. As the outside option changes, so does the relative bargaining strength of each side; organizations must respond accordingly. Organizations use a combination of direct incentives (the amount offered) and indirect incentives (additional time to search for a better outside option). For instance, as well as choosing how much to offer, the organization sets conditions on when the offer expires. Rather than a take-it-or-leave-it offer, the agent may be given the opportunity to explore the outside option before deciding; more time provides better option value.

This paper studies the optimal design of dynamic negotiation policies with a changing outside option. What combination of direct and indirect incentives does the firm use? How does the firm's policy depend on the agent's outside option? Does the policy apply more pressure to accept when the outside option is high or low? What are the implications for the outcomes of these negotiations? How does the evolution (stochastic process) of the outside option affect these outcomes?

*University of Chicago Booth School of Business (email: andrew.mcclellan@chicagobooth.edu). Jeff Ely was the coeditor for this article. I would like to thank the coeditor and three anonymous referees for their comments and suggestions. I am especially grateful to Debraj Ray for his invaluable feedback, guidance, and comments. I would also especially like to thank Dhruva Bhaskar for many helpful comments and discussions as well as Simon Board, Eric Budish, Isa Chaves, Jack Fanning, Emir Kamenica, Rohit Lamba, Elliot Lipnowski, Pietro Ortoleva, Doron Ravid, and Jakub Steiner for useful feedback. This work is supported by the Liew Family Junior Faculty Fellowship at the University of Chicago Booth School of Business.

[†]Go to <https://doi.org/10.1257/aer.20230015> to visit the article page for additional materials and author disclosure statement(s).

We analyze a stylized model to address these questions and apply a dynamic contracting approach to a classic “split-the-pie” problem between a principal and an agent, where the agent’s outside option evolves according to a publicly observable diffusion process. The principal can make an offer at each instance, which the agent can accept or reject, and the principal commits to a negotiation strategy. We characterize the principal’s optimal negotiation strategy. The key insight of the paper is that the principal jointly reduces demands and grants additional decision time to persuade the agent to continue negotiating, highlighting a complementarity between these two negotiation tools.

Optimal negotiations feature a novel nonstationary set of dynamics in which the principal gradually adjusts his demands as well as the length of time the agent is given to consider the offer. Delay and negotiation breakdowns are both prevalent and efficient: rather than make an exploding offer, the principal may give the agent time to explore this outside option. This additional time allows the principal to make a higher demand if the agent’s outside option falls while, by giving the agent option value, ensuring the agent prefers negotiating over taking his outside option. In essence, the principal insures the agent against decreases in the outside option.

The optimal policy takes a simple form. There is an upper *breakdown* threshold, such that if the outside option ever crosses it, the agent leaves and takes the outside option. At any date, the principal’s negotiation strategy is characterized by the current outside option and the highest realized outside option to date, the maximum of the agent’s past outside options. Whenever the agent receives a higher outside option than in the past, the principal makes a better offer and sets a lower *split* threshold. As long as the agent’s outside option stays above the split threshold, the offer remains on the table, and the agent can explore his outside option. However, if the outside option crosses the split threshold, the offer becomes a take-it-or-leave-it one, and the agent immediately accepts it. The placement of the split threshold captures how much pressure is placed on the agent; the higher it is, the less expected time there is for the agent to wait for a better outside option to arrive.

The offer and the split threshold change whenever the outside option reaches a new high in order to ensure that the agent does not leave. The principal has two instruments to provide the necessary incentives: direct incentives through a better offer or indirect incentives with a lower threshold and more option value. These instruments are linked, as a better offer only impacts utilities if and when the offer is accepted. The principal possesses a variety of ways to persuade the agent to continue negotiating. For example, the principal could improve the offer and set a tighter threshold or vice versa. In fact, we find that as the maximum of the outside option increases, the principal gradually improves the offer while also lowering the split threshold to provide more option value. Therefore, direct and indirect incentives work as complements rather than substitutes.

These dynamics match those of real-world negotiations. Firms often give workers time to consider a job offer and adjust both the amount of time as well as the details of the offer in response to offers from competing firms. Gradual decreases in demands, delay, and negotiation breakdowns are also commonly observed.¹ In

¹ Backus et al. (2020, p.1348) empirically document such features in buyer-seller negotiations on eBay and conjecture that dynamic outside options may play a role: “There is no reason to expect that the outside options of

addition to showing how outside options can generate these qualitative features of real negotiations, our model yields new predictions; for example, agents in a better initial bargaining position (i.e., a higher outside option) will have longer delay before an agreement is reached.

We use the model to study comparative statics on the value of negotiations relative to changes in the evolution of the outside option. While an increase in A 's outside option lowers the principal's continuation value (as one might expect), we find that the principal's value from the relationship is *increasing* in the drift of the agent's outside option. This result may be surprising at first glance; a higher drift improves the agent's expected outside option in the future, and an increase in the outside option lowers the principal's value of negotiating. To gain some intuition, consider the effect of an increase in the drift holding fixed the negotiation strategy. In addition to increasing the future outside option, which hurts the principal, an increase in the drift makes exploring the outside option more valuable to the agent, increasing his value from negotiating. The principal can then modify his strategy to extract this additional surplus from the agent, which we show dominates the negative effect of higher future outside options. A similar comparative static holds on the volatility of the agent's outside option: under a weak additional assumption on the outside option, the principal's expected utility is increasing in the volatility of the outside option. The intuition for this is best captured when the drift is zero, as higher volatility "speeds up" the game, reducing delay time. The comparative statics extend to aspects of the optimal strategy; the principal's demand function is increasing in the drift and volatility.

These comparative statics give insights into the principal's preferences over different negotiation environments. All else equal, firms will prefer to bargain with "superstar" workers who are likely to receive better outside options in the future. Similarly, firms will prefer workers who have a greater uncertainty or learn more quickly about their outside options. Firms may then find it optimal to take actions during negotiations that increase a worker's future outside options (e.g., a firm negotiating with a current employee may give the worker time to interview with competing firms or provide a positive reference for the worker).

The paper proceeds as follows. In Section I, we specify the baseline model. In Section II we illustrate the value of delay and changing offers, and in Section III, we solve for the optimal contract and discuss its properties. Comparative statics are provided in Section IV. All proofs are relegated to the Appendix or the online Appendix.

Related Literature.—Our work contributes to the dynamic contracting literature. Some features of the optimal negotiation strategy in our model match those found in other dynamic contracting settings. Back-loading of incentives arises in our model as well as in general principal-agent models (Ray 2002; Bird and Frug 2022), the design of self-enforcing contracts (Thomas and Worrall 1988), apprenticeship dynamics (Fudenberg and Rayo 2019), project selection (Forand and Zápal 2020),

two bargaining parties will stay constant even during the short window of negotiation over which we observe them (and outside options of negotiators off the eBay platform may be changing as well). In this light, dynamic outside options may be an additional explanation for the delayed agreement and delayed disagreement."

the design of approval rules to incentivize experimentation (McClellan 2022), and optimal promotion schemes (Durandard 2023). Our solution also features rigidity in the actions taken, which is present in many of these papers.² The use of “on-path breakdowns” that we find mirrors results in Fuchs and Lippi (2006), who show how breakdowns may be efficient when sustaining the relationship after a sufficiently large “shock” to the environment is too costly.³

Our model features dynamic contracting in real-option environments, which has been studied by papers such as DeMarzo and Sannikov (2017) and Gryglewicz and Hartman-Glaser (2020). Unlike our model, these papers focus on optimal contract design using transfers when facing agent moral hazard. DeMarzo and Sannikov (2017) consider a changing outside option and show the optimal contract uses a changing termination threshold at which the agent is forced to take his outside option. Termination is *ex post* inefficient and serves to punish the agent as a deterrent for shirking. Although an evolving threshold also arises in our model, the purpose of this threshold and its evolution is significantly different.

An important difference between our paper and the previous literature is the set of tools available to the principal in the optimal contract. Our principal can incentivize the agent either directly by offering an increased share of the pie or indirectly by offering additional option value via longer time to explore the outside option. The previous literature has focused on direct incentives through a choice of action (e.g., transfers) by the principal when actions are separable over time; namely, the costs and benefits of an action today are independent of previous actions.⁴ In our model, the actions of *how and when* to split the surplus are not separable; stopping is an irreversible action, and, because the share of the pie offered to the agent only affects players’ utilities if and when a split is agreed to, the impact of increasing the offer (i.e., giving more direct incentives) depends on the expected length of search time before the offer is accepted (i.e., how indirect incentives are provided). This difference brings new challenges to solving for the optimal contract, and intuitions behind the structure of optimal contracts in the previous literature do not apply in our setting. Our model allows us to derive new insights into the optimal structure and interaction of two natural components of negotiation strategies, how much to demand and how much pressure to apply, as well as novel comparative statics on how a “better” outside-option process for the agent benefits the principal.

Our setting is quite close to those studied in the bargaining literature. Two important features of our model—the outside option and the stochasticity of the

²Similar dynamics may arise in settings without commitment; Dixit, Grossman, and Gul (2000) study policy selection by parties under political power fluctuations and find a similar rigidity in efficient equilibria.

³Termination in dynamic contracts may be efficient for other reasons as well, such as providing punishments to agents protected by limited liability or as a way of saving on costs when the agent becomes too expensive to motivate (Spear and Wang 2005).

⁴Two exceptions are McClellan (2022), who focuses on the design of stopping rules alone to provide incentives to perform experimentation, and Durandard (2023), who focuses on the design of experimentation and promotion rules in a multiagent setting. Our model differs in a number of important ways. We allow the principal to choose when to stop and how much demand upon stopping, giving him a richer set of tools to incentivize the agent. Preferences also differ in our model; e.g., McClellan (2022) studies an environment where the agent prefers to stop sooner than the principal, while in our model the agent will prefer more delay than the principal.

environment—have been well studied in this literature.⁵ However, our connection to this literature is less direct; our paper, by allowing commitment, focuses on studying different forces than those in the bargaining literature, where the inability to commit to future offers is a crucial factor for determining equilibrium outcomes.⁶

Finally, we are related to papers looking at how to optimally set offer deadlines for fixed demands when the agent can search for a better outside option and no agreement can be reached after the deadline has passed (Lippman and Mamer 2012; Zorc and Tsetlin 2020; Hu and Tang 2021). This literature has focused on using deadlines to deter search by the agent, while our paper shows how delay can be used to benefit the principal by providing option value to the agent. Our paper also differs by allowing for general contracts that can condition on changes in the outside option. This general contracting space allows us to explore how the principal's negotiation strategy evolves over time and gives us new insights into the interaction between how much to demand and how much search time to grant the agent.

I. Model

Two players, a principal P and agent A , must decide how to split a pie of size one. Time is continuous, with an infinite horizon, and discounted by both players at rate $r > 0$. At each moment in time, P makes a demand for a share of the pie, which A can either accept or reject (we will sometimes refer to the corresponding share for A as the offer). If A accepts, the game ends, and each player i receives utility $u_i(z_i)$ from the share $z_i \in [0, 1]$ they receive, where $z_P + z_A = 1$. We assume each u_i is twice differentiable, concave, and strictly increasing with $u_i(0) = 0$.⁷

Each player has an outside option they can take at any time, in which case the game immediately ends with both players receiving a payoff equal to their outside option. P 's outside option is fixed, with a value of zero.⁸ A 's outside option evolves over time, with a value at time t of Y_t , where Y is a publicly observable diffusion process with initial value Y_0 solving the stochastic differential equation

$$dY_t = \mu(Y_t)dt + \sigma(Y_t)dB_t,$$

where B is a Brownian motion on a canonical probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ with continuously differentiable μ and σ , twice so for σ . We assume there exist

⁵The role of outside options in bargaining goes back to Nash (1950), whereas recent papers studying changing bargaining environments have looked at the impact of newly arriving players (Fuchs and Skrzypacz 2010; Chaves 2019), transparency of outside options (Hwang and Li 2017), the arrival of information about a seller's type (Daley and Green 2020), changing costs of supplying a good (Ortner 2017), reputation about the outside option (Compte and Jehiel 2002; Lee and Liu 2013), and a changing size of the pie to be split (Merlo and Wilson 1995).

⁶A previous version of the paper also studied a discrete-time bargaining game without commitment and showed that, as the period length shrinks to zero, there exists an equilibrium generating the same utility for the principal as under commitment. This game possessed a multiplicity of equilibria, some of which are used in the construction of an equilibrium converging to our commitment benchmark; such forces have been previously observed in papers such as Ausubel and Deneckere (1989).

⁷We take increasing/decreasing/concave to be weakly increasing/decreasing/concave and will specify when such functions are strictly so.

⁸A previous version of the paper allowed for $u_i(0) > 0$ and P 's outside option to have a strictly positive value and showed that our main results are identical.

$0 \leq \underline{Y} < \bar{Y}$ such that $Y_t \in (\underline{Y}, \bar{Y})$ for all t and $\sigma(y) > 0$ for all $y \in (\underline{Y}, \bar{Y})$.⁹ To avoid trivial versions of the model, we assume $u_A(1) > \underline{Y}$.

P can commit to how he makes demands and when to take his outside option, which we call a *contract*,¹⁰ and can condition his actions at each time t on his past actions and the observed path of A 's outside option $\{Y_s\}_{s \in [0, t]}$. Without loss, we focus on contracts in which P chooses a stopping time at which to either make a take-it-or-leave-it demand or take his outside option.

DEFINITION 1: A **contract** consists of $\mathcal{F}_t$ -measurable functions $(\tau, d_\tau, \alpha_\tau)$, where

- (i) τ is a stopping time that gives the time at which the game ends.
- (ii) $d_\tau \in \{0, 1\}$ is a decision rule that equals 1 if and only if a split is made at τ .
- (iii) $\alpha_\tau \in [0, 1]$ gives P 's share of the surplus if a split is made at time τ .

P 's expected utility from a contract $(\tau, d_\tau, \alpha_\tau)$ is

$$J(\tau, d_\tau, \alpha_\tau) := E_{Y_0} [e^{-r\tau} d_\tau u_P(\alpha_\tau)],$$

and A 's expected utility is

$$V(\tau, d_\tau, \alpha_\tau) := E_{Y_0} [e^{-r\tau} (d_\tau [u_A(1 - \alpha_\tau) - Y_\tau] + Y_\tau)],$$

where E_y denotes the expectation given $Y_0 = y$; we use similar notation for $\mathbb{P}$. For notational convenience, we will drop dependence on Y_0 in E_{Y_0} where it causes no confusion.

Because A has the option to take his outside option at any time, P 's choice of a contract must respect individual rationality constraints that ensure A finds continuing the negotiation preferable to taking his outside option.

DEFINITION 2: $(\tau, d_\tau, \alpha_\tau)$ is **dynamically individually rational (DIR)** if for every $t \geq 0$ and history up to t , A 's continuation value at t is weakly greater than Y_t .

Without loss, we focus on contracts in which A never takes his outside option.¹¹ A contract is optimal if it solves

$$(1) \quad \sup_{(\tau, d_\tau, \alpha_\tau)} J(\tau, d_\tau, \alpha_\tau)$$

such that $(\tau, d_\tau, \alpha_\tau)$ is DIR.

⁹Thus, $\underline{Y}$ and $\bar{Y}$ are "natural" boundaries that are never reached in finite time.

¹⁰We need not take the term "contract" to necessarily mean a legal court-enforceable contract. We use the term simply to highlight that P is committed to the specified strategy.

¹¹Replacing any instance of A taking his outside option with P doing so does not change players' payoffs.

Finally, we place one relatively weak assumption on the outside option process.

ASSUMPTION 1: $e^{-rt}Y_t$ is a supermartingale.

Assumption 1 is equivalent to $ry \geq \mu(y)$ for all $y \in (\underline{Y}, \bar{Y})$. By an application of Doob's optional-stopping theorem, Assumption 1 implies $E_y[e^{-r\tau}Y_\tau] \leq y$ for any τ . Thus, if A knows he will take his outside option for sure in the future, then he is better-off taking it today. We say $e^{-rt}Y_t$ is a *strict supermartingale* if $ry > \mu(y)$ for all $y \in (\underline{Y}, \bar{Y})$.¹² Any $e^{-rt}Y_t$ satisfying Assumption 1 can be made a strict supermartingale by increasing r slightly.

Assumption 1 is a natural property to impose on A 's outside option. Suppose $e^{-rt}Y_t$ is not a supermartingale but that, after walking away from the negotiation, A is able to let his outside option continue to evolve before taking it. Let $v(Y_t) := \sup_\tau E_{Y_t}[e^{-r\tau}Y_\tau]$ be A 's optimized value at t of choosing when to take his outside option conditional on the value of Y_t . By standard optimal stopping results, $e^{-rt}v(Y_t)$ is a supermartingale.¹³ $v(Y_t)$ gives A 's value of walking away at time t ; that is, it is his "true" outside option. Replacing Y_t with $v(Y_t)$, our model will satisfy Assumption 1.

Comments on the Model.—

Commitment: We assume P can commit to his negotiating strategy. This assumption is reasonable, for example, when P represents a large organization that can design the rules by which its representatives negotiate; e.g., a firm may delegate negotiating to a manager who is restricted to not change the current offer unless the worker shows he has received a sufficiently attractive outside offer from another firm. In addition, organizations have reputational incentives when they negotiate with many agents over time that allow them to credibly commit to their offer strategy. If workers learn the firm has, in past negotiations, reneged on the offer specified in the contract, they may lose trust that the firm will stick to its promised offers and avoid negotiating with the firm. The costs of lost future negotiations, if large enough, will deter P from deviating from the contract.¹⁴

Observability: We also assume A 's outside option is observable.¹⁵ In sovereign-debt or merger-agreement negotiations, the evolution of the outside option is driven by changes in economic fundamentals that are observable to both parties. In firm-worker negotiations, the worker can disclose to their current firm

¹²This implies for each τ with $\mathbb{P}_y(\tau > 0) > 0$, we have $E_y[e^{-r\tau}Y_\tau] < y$.

¹³This will not necessarily be a strict supermartingale: if $v(y) > y$ on some interval $[y_1, y_2]$, then $e^{-rt}v(Y_t)$ will be a martingale on that interval. However, we can always make it strict by adding arbitrarily small flow costs of continuing to negotiate.

¹⁴Deviating from the contract might come at other costs besides the loss of reputation in future negotiations. Empirical evidence shows that, in firm-worker negotiations, practices by a firm while negotiating that are perceived as "unjust" (as reneging on a promised offer would be) can lead to a desire for retribution among workers (Skarlicki and Folger 1997), as well as low organizational commitment and high turnover (Ferguson, Moye, and Friedman 2008). Such effects lower the value of hiring the worker, namely, decrease the size of the pie to be split.

¹⁵Similar assumptions on the observability of outside options are made in previous dynamic contracting papers, such as Thomas and Worrall (1988).

any written offers received from competing firms.¹⁶ Alternatively, the outside option might correspond to a “market” value for the worker outside of his current firm (and with whom he is negotiating over a new contract and whose value of the worker is fixed), with changes in the outside option coming from “career progress,” as information about the worker’s abilities or acquisition of new skills changes his market value. Such information can be observed by the worker’s current firm as well as the outside market.¹⁷

II. Value of Delay and Dynamic Offers

We begin our analysis by highlighting two incentive tools that play a key role in the optimal policy: offering *A option value* and *dynamically increasing offers*. First, *P* can make an offer that remains on the table as long as *A*’s outside option is sufficiently high. This provides *A* indirect incentives through option value to wait and see if the outside option improves, while insuring him against the downside; if the outside option falls, *A* can agree to a split that gives him more utility than his outside option. In exchange for some option value, *P* can demand more than he could with a take-it-or-leave-it offer, which offers no option value.

To illustrate this, consider a simplified two-period version of the model without discounting. Suppose at $t = 0$, *A*’s outside option is $y = u_A(1)$ (the value of the entire pie) and at $t = 1$, for some $\epsilon \in (0, u_A(1))$, the outside option is $y + \epsilon$ with probability q , and $y - \epsilon$ with probability $1 - q$, with $q \in (0, \frac{1}{2})$. The only take-it-or-leave-it offer at $t = 0$ that *A* will accept is the entire pie, which gives *P* a utility of $u_P(0) = 0$. In contrast, an offer of $1 - \Delta$ (for some small Δ), which remains on the table until for both periods, is accepted at $t = 1$ if the outside option falls to $y - \epsilon$, and, moreover, provides *P* a payoff of $(1 - q)u_P(\Delta) > 0$. *A* is willing to delay knowing that if his outside option does fall, he will not suffer the full amount of the decrease, receiving $u_A(1 - \Delta)$ instead of $y - \epsilon$ (see Figure 1).

In this example, because *A*’s outside option is high to start with, a take-it-or-leave-it offer cannot help *P*, whereas indirect incentives can. This idea holds quite generally. It has a natural interpretation for applications. Firms may find it very expensive to immediately hire a worker with a high outside option and so instead offer the worker a lower offer with a longer deadline, which allows the worker to explore outside offers while being insured against the event that nothing better arrives.

Returning to the infinite-horizon problem, the first point tells us that it can be optimal to offer *A* less than the entire pie but give option value by keeping the offer on the table for some time. If *P* commits to keep his demands constant, *A* will agree to a split if he becomes sufficiently pessimistic about the chance of receiving better outside options, namely, when the outside option hits some lower threshold. But how should *P* respond if the outside option goes up? If *P* never adjusts the offer, *A* will leave when his outside option is sufficiently good, namely, when it hits some upper threshold.

¹⁶ As we will see, *A* has no incentive to conceal an increase of his outside option in the optimal contract.

¹⁷ A similar assumption of publicly observable news is made in Daley and Green (2020), who study one-sided incomplete information bargaining when commitment is not possible.

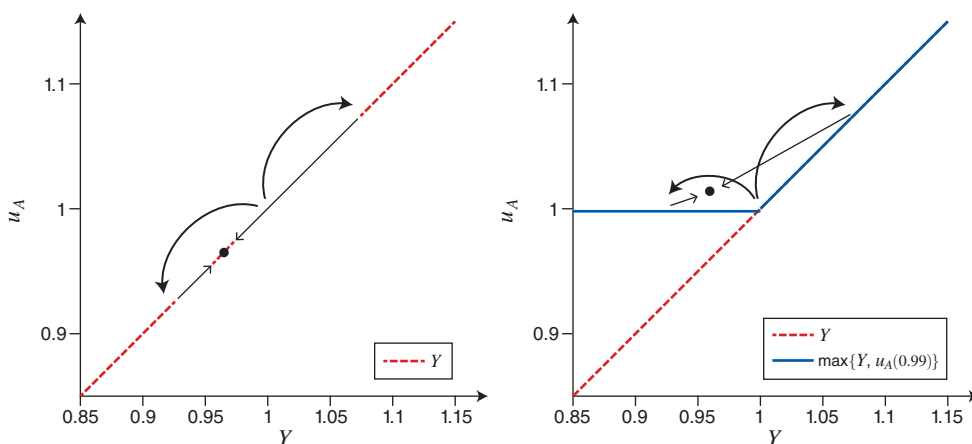


FIGURE 1

Notes: Curved arrows indicate the movement of the outside option. The black dot gives the expected value of waiting to take the outside option in the left graph and waiting to decide between a 0.99 share of the pie or the outside option.

Can it be optimal to keep demands constant? Upon reaching this upper threshold at which A will leave, the current offer on the table gives A strictly less than the entire pie, so there is room to increase the offer. Moreover, if A leaves, P gets nothing. Suppose, instead, upon reaching this upper threshold, P increases the share of the pie offered to A (but still offers less than the entire pie), making A willing to wait a bit longer and continue negotiating.¹⁸ This, in turn, gives rise to the possibility that the outside option will fall enough that A accepts the new offer. The change of offer increases the value of negotiating to A and gives P a positive expected payoff, which is an improvement compared to sticking with the initial offer. In general, whenever A is about to leave, P can always benefit by increasing the value of negotiating for A through direct or indirect benefits and will do so when feasible. In other words, dynamically changing offers can improve upon stationary policies.

III. Optimal Contract

Following the logic of the preceding discussion, an optimal contract will involve improving A 's continuation value by continuing to negotiate whenever A is tempted to leave. Implementing this idea requires us to understand *when* A is tempted to walk away and exactly *how* to optimally increase his continuation value. In general, this is a complicated question for arbitrary contracts, as A 's continuation value can depend on histories in complex ways. In addition, there are numerous ways in which P can increase A 's continuation value, e.g., lowering demands for some time before

¹⁸ Because A is choosing when to take the offer or his outside option, this stationary contract is *DIR*. This implies A must be indifferent between continuing to negotiate and taking the outside option at the upper threshold, so any increase in P 's offer will make A strictly prefer to continue negotiating. The assumption of *DIR* is crucial for the necessity of nonstationary contracts; it is straightforward to show that the optimal contract will be stationary if we replace *DIR* with an ex ante participation constraint $V(\tau, d_\tau, \alpha_\tau) \geq Y_0$.

raising them in the future or changing the stopping time at which an agreement is reached. Furthermore, changes to demands and stopping times will interact, as the impact of changes in demand on players' expected utilities depends on when and if a split is reached.

Despite such complexities, our main result shows that the optimal contract takes a relatively simple form that depends only on two state variables, Y_t and the maximum of the outside option process $M_t = \max_{s \in [0, t]} Y_s$. The structure of this contract, characterized in Theorem 1, sheds insights into how P optimally persuades A to continue negotiations and shows how two natural aspects of any negotiation strategy, namely what demands to make and how much pressure to apply, evolve over time and highlights a complementarity between them.

THEOREM 1: *For some decreasing functions S^* and α^* and constant $\bar{R}^* < \bar{Y}$, there is an optimal contract $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ that reaches a split whenever $Y_t \leq S^*(M_t)$ with a demand of $\alpha^*(M_t)$ and takes the outside option whenever $Y_t \geq \bar{R}^*$:*

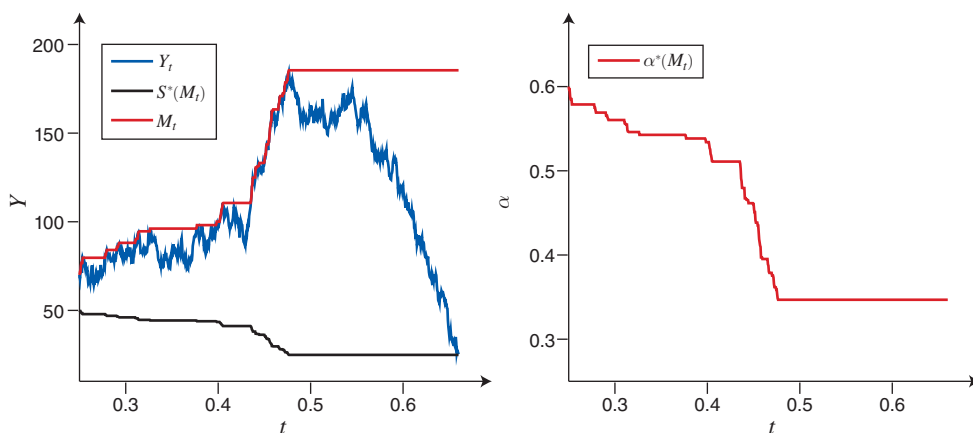
$$\tau^* = \inf\{t : Y_t \notin (S^*(M_t), \bar{R}^*)\}, \quad d_{\tau^*}^* = \mathbf{1}\{Y_{\tau^*} < \bar{R}^*\}, \quad \alpha_{\tau^*}^* = \alpha^*(M_{\tau^*}).$$

α^* is not constant. S^* is constant if $u_i(z) = z$ for both $i \in \{P, A\}$ and is strictly decreasing if $e^{-rt} Y_t$ is a strict supermartingale and either u_P or u_A is strictly concave.

The optimal contract is characterized by a demand function $\alpha^*(M_t)$, a split threshold $S^*(M_t)$, and a breakdown threshold $\bar{R}^*$. The location of the split threshold $S^*(M_t)$ gives the amount of *pressure* being placed on A and can be thought of as (roughly) a *deadline* for the offer: the higher S^* is, the less time A has to explore his outside option before being forced to make a decision. That α^* and S^* are decreasing while M_t is increasing over time implies P decreases his demands and pressure over the course of negotiation; that is, they are complementary tools for incentivizing A to continuous negotiating, strictly complementary if utilities are strictly concave and Assumption 1 is strict.¹⁹ The breakdown threshold $\bar{R}^*$ is given by the point at which A would choose to take his outside option in A 's first-best contract (which implies that $\bar{R}^*$ is history independent); that is, P continues to concede demands and pressure to A until it is not possible for him to increase A 's value of negotiating any further, at which point he lets A walk away. Figure 2 illustrates the optimal contract over a sample path of Y , showing how the stochasticity of when M_t increases leads to fixed demands for long periods of time followed by quick bursts of concessions when M_t rises.²⁰

¹⁹ While we require a strict supermartingale condition to ensure that S^* is strictly decreasing everywhere, one can show that S^* is strictly decreasing somewhere if u_P or u_A is strictly concave.

²⁰ While we have, without loss, assumed that P makes a take-it-or-leave-it offer at τ^* (with no offers made prior to τ^*), this is not necessary. Because $\alpha^*(M_t)$ is decreasing, P would always be better-off if A were to accept the split at (Y_t, M_t) with $Y_t > S^*(M_t)$. Optimality of the contract then implies that A must prefer to wait until τ^* . Thus, the optimal contract can be equivalently implemented by P making the offer $\alpha^*(M_t)$ at each date $t < \tau^*$, with A rejecting the offer at each such t .

FIGURE 2. THE EVOLUTION OF THE OPTIMAL CONTRACT OVER A SAMPLE PATH OF Y

Role of M_t .—A key feature of the optimal contract is that the demands and split threshold depend only on M_t even though it is not directly payoff relevant²¹—as long as M_t is constant, the offer on the table and its deadline S^* do not change—while Y_t determines when P threatens to take the offer off the table. The offer is set so that upon reaching a high outside option that increases M_t , A will take his outside option if the current offer is not revised; that is, P only improves the offer or deadline when A can credibly threaten to take his outside option (namely, when *DIR* binds), rather than changing the offer initially. If M_t increases, P is forced to improve the terms. Thus, M_t captures the past promises made to A to persuade him to continue negotiating. Upon the outside option falling (i.e., $Y_t < M_t$), *DIR* slackens. Nevertheless, P must continue to keep the improved offer on the table so that his past promises to A are credible. As increases in M_t result in better offers (which are held fixed as long as M_t is constant), this raises A 's continuation value at all $Y_t < M_t$ relative to previous offers.

Given the flexibility of the optimal contract, there are many ways in which P could deliver on his past promises (e.g., lower demands today before raising them in the future). The intuition for stationarity of the demand and split threshold in M_t comes from a smoothing motive that induces P to keep both constant when *DIR* is slack (namely, as long as M_t is constant). While smoothing P 's demands is intuitive by the concavity of u_P and u_A , a similar concavity intuition does not apply to stopping times; even fixing attention to threshold stopping times, players' payoffs over different thresholds are not globally concave. However, we know from single decision-maker optimal stopping problems that using a fixed stopping threshold is generally optimal. A similar force makes it optimal to keep the split threshold fixed until forced to change it when M_t increases.²²

²¹That is, the set of continuation values that can be delivered to players at some date t depends only on Y_t and not M_t .

²²Roughly speaking, the value of continued search at dates s and t with $(Y_s, M_s) = (Y_t, M_t)$ is the same as the payoff-relevant variable (Y) and past promises (M) are the same, in which case the decision of whether to stop will be the same at s and t .

Complementarity of Direct and Indirect Incentives.—We have determined when P changes contract terms; as long as negotiations continue, for each y , A 's continuation value must improve at $(Y_t, M_t) = (y, m)$ whenever m increases, through direct (smaller demand α^*) or indirect (longer deadline, lower S^*) incentives or both. Both tools are costly for P ; clearly, a smaller demand directly decreases the value of an agreement, while lowering S^* entails more delay and a greater probability that M increases again before an agreement is reached. How should P use these two instruments? P could increase A 's continuation value by decreasing α^* while increasing S^* or vice versa. The fact that α^* and S^* are both decreasing in M_t shows a complementarity between these two dimensions of P 's strategy, strictly so if at least one of u_P or u_A is strictly concave.

To provide some intuition for this complementarity, consider a reduced-form version of P 's problem that captures the relevant trade-off for P . For some $m > Y_0$, P chooses a fixed stopping threshold S and demand α when both P and A 's continuation values at the first instance Y crosses m are fixed to be J_m and V_m , respectively. In addition, P is constrained to deliver a continuation value of W to A .²³ Changes in W in this reduced-form problem (roughly) capture how the optimal contract must change after an increase in M_t , the comparative static of interest.²⁴ For simplicity, we assume $u_P(\alpha) = \alpha$.²⁵

Let $\tau_-(S) := \inf\{t: Y_t \leq S\}$, and define the discounted probabilities of crossing S before m and m before S to be

$$\phi(S) := E\left[e^{-r\tau_-(S)} \mathbf{1}\{\tau_-(S) < \tau_+(m)\}\right], \Phi(S) := E\left[e^{-r\tau_+(m)} \mathbf{1}\{\tau_-(S) > \tau_+(m)\}\right].$$

A 's expected utility for any S and α is then

$$\phi(S) u_A(1 - \alpha) + \Phi(S) V_m.$$

Setting this equal to W gives the promise-keeping constraint, and fixing S and W , the demand is pinned down as $\alpha(S, W) := 1 - u_A^{-1}\left(\frac{W - \Phi(S) V_m}{\phi(S)}\right)$. P 's utility from threshold S is then

$$(2) \quad F(S, W) := \phi(S) \alpha(S, W) + \Phi(S) J_m.$$

As W increases, P 's demands α decreases for each S . Does this reduction in his demands induce P to grant more or less decision time (i.e., lower or raise the split threshold S)?

²³This corresponds to the problem of choosing $S^*(m')$ and $\alpha^*(m')$ for $m' \geq m$ when W is A 's continuation value when $(Y_t, M_t) = (Y_0, m')$ and J_m, V_m are each players' continuation value when $(Y_t, M_t) = (m, m')$. $S^*(m')$ and $\alpha^*(m')$ solve our simplified problem given these values of W, J_m, V_m .

²⁴Here, our simplified problem differs slightly from the full problem, as an increase in M_t will require increasing A 's continuation value at (Y_0, M_t) but will also change continuation values at m . However, the forces captured in this simplified problem will illustrate the driving forces in the full proof of the dynamics of $S^*(\cdot)$. The role of assuming $e^{-rt} Y_t$ is a strict supermartingale in Theorem 1 appears here, as A 's continuation value at (Y_0, M_t) is strictly increasing in M_t if $e^{-rt} Y_t$ is a strict supermartingale, but may be constant if $e^{-rt} Y_t$ is a martingale over some region.

²⁵This assumption is without loss: we can always change the units of the demand to be in terms of P 's utility so that $u_A(1 - \alpha)$ is the utility to A of a demand that delivers α utility to P . This change preserves concavity of u_A .

We first consider the limit case in which u_A is linear with $u_A(1 - \alpha) = 1 - \alpha$. In this case, $F(S, W) = \phi(S) + \Phi(S)(J_m + V_m) - W$. P 's expected utility is separable in S and W , and the optimal choice of S is independent of W and simply maximizes $\phi(S) + \Phi(S)(J_m + V_m)$, which is the players' joint utility. Only the demand is determined by W . That is, if A is risk neutral, utility is perfectly transferable through changes in demands, and choosing S to maximize joint surplus and allocating surplus through demands is the most efficient way to increase A 's utility for any W . In other words, only direct incentives are needed, and the threshold S does not change as W increases, staying fixed at the efficient level.

In contrast, if A has strictly concave utility, $F(S, W)$ is no longer separable in S and W , and another trade-off arises. Concavity of u_A means increases in the offer have a decreasing marginal value for A . In a policy that fixes S at the efficient level, as W rises, P needs to offer increasingly larger amounts. This makes it expensive for P to deliver value only through direct incentives. Instead, P can provide some of the increased continuation value through indirect incentives by moving S away from the efficient threshold toward A 's preferred point, allowing him to reduce the increase in the offer. That is, a decrease in P 's demands at a given S increases the benefit to P of changing S . This slows the rate at which α decreases, reducing the variability of the offer, which is beneficial due to the strict concavity of u_A , so P optimally uses both tools to deliver increases in A 's continuation value.

In principle, moving S toward A 's preferred level could involve an increase or decrease in S , as A 's preferences over the threshold are not monotonic.²⁶ The direction of improvement depends on players' relative preference over the duration of search. When demands are decreasing in M_t , P always prefers to stop as soon as possible; additional search incurs discounting costs and the risk that M_t increases, forcing P to lower his demands. Therefore, $S^*(M_t)$ must always be above A 's preferred level given P 's demand $\alpha^*(M_t)$,²⁷ so changing the threshold in A 's preferred direction leads to a decrease in the threshold; namely, S^* is strictly decreasing.

Additional Discussion.—

Delay and Bargaining Position: The optimal contract gives A a short amount of time to consider the offer (i.e., uses a high S^*) when the maximum over past outside options is low and only extends the length of time A has to consider the offer if A is able to come to P with an improved outside option. Thus, a better initial bargaining position for A —a higher Y_0 —leads to longer delay before any agreement is reached.

Negotiation Breakdowns: Another implication of the optimal contract is that breakdowns (i.e., A taking the outside option) happen with positive probability whenever there is some delay in the optimal contract. Thus, optional negotiations feature the type of breakdowns after delays often seen in real-world negotiations. Our model illustrates the fact that such breakdowns may be efficient. This result follows from the fact that allowing A to explore his outside option creates value only

²⁶For low enough Y , A will prefer to take the split relative to continuing search. Moreover, A 's preferred level of the threshold changes as P 's demand changes.

²⁷If not, both players can be made better-off by increasing the threshold.

if it leads him to take the outside option; namely, delay only has value if it leads to a change in A 's decision to accept a split or take the outside option.²⁸

Initial Value Independence: The proof of Theorem 1 shows A 's continuation value is equal to his outside option when $Y_t = M_t$. Thus, the continuation contract at t with $Y_t = M_t = m$ must maximize P 's continuation value subject to *DIR* and delivering m continuation value to A . This is the same problem solved by the optimal contract when $Y_0 = m$, which is then an optimal continuation contract when $Y_t = M_t = m$. Therefore, the form of the optimal contract is *independent* of the starting Y_0 ; namely, $S^*, \alpha^*, \bar{R}^*$ are the same for all values of Y_0 . This independence relies on the ability of the contract to respond to changes in the outside option; if, for example, we were to restrict P to only choose among stationary policies (i.e., a constant demand and split threshold), the optimal policy *would*, in general, depend on Y_0 .

Renegotiation: Given the history dependence of the optimal contract, we may be concerned that players would have an incentive to renegotiate the contract if continuation play at some history is inefficient. This is not a concern if the contract is *constrained Pareto efficient*, namely, after every history there is no *DIR* contract that weakly increases both players' utilities, strictly so for at least one player. The optimal contract must be constrained Pareto efficient after each history prior to τ ; otherwise, P could modify the continuation contract at dates where the contract is inefficient to increase his own utility, while keeping A 's the same. However, we also need to consider what happens if A were to turn down an offer at τ that the contract calls for him to accept. The optimal contract must then punish A ; it is without loss to assume it delivers to A the lowest continuation value possible, which is equal to A 's outside option. There may be renegotiation concerns if this is done through P taking his outside option. However, we can always hold A 's continuation value to his outside option in a constrained Pareto-efficient way by, upon A turning down an offer at τ , using the optimal contract when $Y_0 = Y_\tau$ as the continuation contract at τ because it gives A a continuation value equal to his outside option, thereby removing concerns about renegotiation of the contract.

IV. Comparative Statics

We now study how P 's value of negotiating depends on the outside-option process. Such comparative statics give us a better understanding of what aspects of the environment drive the value of the negotiation. They are also useful for studying players' preferences over different environments. Given a choice over multiple agents who differ in their outside options, which agent will P choose to bargain with? If players can shape the outside-option process, how should they do so?

We first consider the effect of changing the current outside option. Unsurprisingly, P 's continuation value at t is decreasing in Y_t .²⁹ The intuition is simple: increasing Y_t

²⁸Our model can capture a special case of the standard durable goods monopolist, namely, a known buyer value for the good, by taking $\sigma(\cdot)$ to zero, so that A 's outside option is constant over time. In this case, because there is no option value generated by delay, the optimal contract stops immediately, with either a split realized or the outside option taken immediately.

²⁹This result is a straightforward implication of the fact that S^* and α^* are both decreasing.

increases both the delay before an agreement is reached and the probability P will have to lower his future demands.

The more interesting comparative statics arise when we look at changes in how the outside option evolves. Formally, we compare negotiations when A 's outside option is Y to those when we replace A 's outside option Y with another diffusion process $\hat{Y}$ on $(\underline{Y}, \bar{Y})$ that solves the stochastic differential equation

$$d\hat{Y}_t = \hat{\mu}(\hat{Y}_t)dt + \hat{\sigma}(\hat{Y}_t)dB_t, \quad \hat{Y}_0 = Y_0,$$

where $\hat{\mu}$ and $\hat{\sigma}$ are continuously differentiable, twice so for $\hat{\sigma}$, and $\hat{\sigma}$ is strictly positive. Throughout, we assume $\hat{Y}$ also satisfies Assumption 1. Thus, the optimal contract under $\hat{Y}$ has an analogous form to that in Theorem 1. We let $\alpha^*(\cdot)$ and $\hat{\alpha}^*(\cdot)$ denote P 's demand function in the optimal contract under Y and $\hat{Y}$, respectively. For m above the breakdown threshold $\bar{R}^*$, we take $\alpha^*(m) = 0$ without loss and similarly for $\hat{\alpha}^*$.

Consider an increase in the drift: $\hat{\mu} > \mu$ and $\hat{\sigma} = \sigma$. Because such a change will lead to an increase in the future values of the outside option, and an increase in the outside option reduces P 's continuation value, a natural conjecture is that an increase in μ will also reduce P 's value of the negotiation; namely, his expected utility from the optimal contract in (1). Proposition 1 shows that, in fact, the comparative static goes in the *opposite* direction. This result illustrates an important distinction between the *current level* of outside option and the *expected level* of the outside option tomorrow: increasing the current level hurts P , whereas increasing the expected level tomorrow benefits P .

PROPOSITION 1: *If $\hat{\mu} > \mu$ and $\hat{\sigma} = \sigma$, then P 's value of the negotiation is higher under $\hat{Y}$ than Y and $\hat{\alpha}^* \geq \alpha^*$.*

To build some intuition for Proposition 1, consider an increase in μ holding the optimal contract fixed. An increase in μ increases the expected outside option tomorrow, which reduces P 's expected utility today. However, search now becomes more valuable to A , slackening *DIR*. P can then modify the contract to extract additional surplus from A without violating *DIR*. Proposition 1 shows this second force is always dominant, leading to a larger value for P .

Intuitively, increasing μ makes the "option-value surplus" larger. Consider a social planner's problem of maximizing some weighted sum of P and A 's utility, ignoring the *DIR*. The social planner's continuation value is always weakly increasing in the outside option, so an increase in μ will increase the value of the social planner's problem. Our result highlights that, even when adding in the *DIR*, an increase in μ still increases the total surplus, and P is able to extract this additional surplus.

However, an increase in Y_0 can also increase the total surplus in the social planner's problem. Why does an increase in the total surplus through an increase in μ reverse the sign of the comparative static on P 's expected utility when compared with an increase in Y_0 ? The difference comes from how the increase in value is allocated to players. The additional value created by an increase in Y_0 is already given entirely to A : due to *DIR*, P is forced to give A a larger continuation value at

$t = 0$ and is not able to extract the additional surplus created. By contrast, when μ increases, A 's participation constraint at $t = 0$ continues to bind (giving him the same expected utility under μ and $\hat{\mu}$), and P alone benefits from the increase in surplus. The key difference is that an increase in μ does not improve A 's position today.

Proposition 1 also shows $\hat{\alpha}^* \geq \alpha^*$; that is, P makes *higher* demands when A 's outside option has a higher drift. This result does not immediately follow from the fact that P 's expected utility under $\hat{Y}$ is higher than under Y . P has multiple ways to change the contract to increase his utility; for example, he can increase his expected utility by raising the split threshold while decreasing his demands. Our result shows increasing demands are an optimal way to extract the additional surplus.

Proposition 1 has implications for what type of agents P would choose to bargain with. An agent with a higher drift expects to receive high outside options in the future (a "superstar"). Even when holding fixed the size of the pie for different types of agents, our result shows P will prefer to bargain with the superstar, despite knowing the probability he will not be able to reach an agreement is higher. Searching for better outside options, although more profitable for A when the drift is higher, comes at the risk of a decrease in the outside option. The superstar is willing to take the additional search time to explore his outside option at the cost of accepting a lower share of the pie when his outside option falls.

This result also implies P benefits from actions that make A 's expected outside option tomorrow better. Consider a firm P and worker A negotiating over a new contract upon the expiration of an old one. P can affect how A 's outside option will evolve by permitting the worker to take time to interview with other firms or giving a positive reference letter. Our results highlight how P benefits from such actions even though they may increase the probability that P is forced to increase his wage offer or that A accepts an outside offer.

Next, we look at the effect of changes in σ . We focus on the case where $\mu = \hat{\mu} \leq 0$ so that Y is a supermartingale. A special case is when $\mu = 0$, namely, when Y is a martingale, which naturally arises in many economic settings, such as when the changes in the outside option are driven by the arrival of new information.³⁰ Our next result shows that, under this slightly stronger supermartingale condition, P benefits from an increase in the volatility of the outside option.

PROPOSITION 2: *If $\hat{\sigma} > \sigma$ and $\mu = \hat{\mu} \leq 0$, then P 's value of the negotiation under $\hat{Y}$ is higher than under Y and $\hat{\alpha}^* \geq \alpha^*$.*

The intuition for the proof is simple in the case in which Y is a martingale. Holding the optimal contract fixed, increasing σ reduces the amount of delay by "speeding up" the outside-option process without changing the probability of an increase or decrease in the outside option, thereby benefiting both parties. More generally, we show P 's continuation value is convex in the outside option when $\mu \leq 0$ and that convexity is a sufficient condition for P 's value to increase in σ . The restriction to $\mu \leq 0$ is sufficient but not necessary for P 's value of the negotiation to be

³⁰For example, suppose A 's "true" outside option is $\theta \in \{0, 1\}$ but is unknown to both players. Players observe a signal $dZ_t = \theta dt + \tilde{\sigma} dB_t$, where B is a Brownian motion. Letting $Y_t \in (0, 1)$ be the expected value of θ at time t , standard results show $dY_t = \sigma(Y_t)dB_t$, where B is a standard Brownian motion and $\sigma(y) = \tilde{\sigma}^{-1}y(1 - y)$.

increasing in σ . It is straightforward to show P 's optimal contract when $\sigma = 0$ will have no delay and must either take the outside option or a split immediately, even if $\mu > 0$. Because this same outcome is feasible under any outside-option process, P 's expected utility must be weakly higher for $\sigma > 0$ than $\sigma = 0$.

In the context of firm-worker negotiations, a higher σ can be thought of as corresponding to a worker with higher uncertainty about the value of outside employment (e.g., workers who are uncertain about how outside firms will value their abilities or skills). Proposition 2 implies P prefers to bargain with workers who have a higher degree of uncertainty about their outside options. The analogy of the optimal contract providing insurance against the possibility of a decrease in the outside option is apt here; higher uncertainty regarding A 's outside option allows P to charge a higher "premium," namely, a higher demand α^* .

V. Conclusion

In this paper, we study the design of optimal negotiating strategies when one player's outside option may change over time and show it generates a rich set of dynamics. The committed party gradually decreases their demands and the pressure on the other party over time, leading to periods of intransigence followed by quick spurts of concession, reminiscent of haggling. By bringing a dynamic contracting approach to a classic bargaining setting, we are able to study natural situations in which one party has some commitment power and provide new insights into how various dimensions on the negotiation strategy interact. Our model shows how natural features of real-world negotiations such as delay, gradual concessions, and breakdowns can arise in an optimal strategy.

Other extensions remain for future work.³¹ Allowing the principal's outside option to evolve as well could bring new dynamics into the structure of the optimal contract. Incorporating private information about outside options is another natural extension. While we expect that many of the same forces identified in this paper would still play an important role, it would be interesting to explore how these dynamics change in the context of private information.

APPENDIX A. PROOF OF THEOREM 1

We assume throughout either u_A or u_P is strictly concave and both have bounded first and second derivatives. The optimal contract in environments that do not satisfy these assumptions can be found by taking the limit of optimal contracts in a sequence of environments satisfying these assumptions.

We first study a discrete-time approximation of P 's problem in (1) with period length $\Delta > 0$, where each period is discounted by $e^{-r\Delta}$.³² Let $\sigma_0 = \max_{y \in (Y, \bar{Y})} \sigma(y)$ and $\epsilon = \sigma_0 \sqrt{\Delta}$. We define a random walk X for $t \in \{0, \Delta, \dots\}$ to approximate Y

³¹ In a previous version of the paper, we showed that the structure of the optimal contract is identical in a number of variants of the model, such as adding transfers; using alternative outside option processes allowing for jumps in the outside option; and a version of the model where the size of the pie changes over time, while outside options are fixed.

³² Solving the discrete-time problem, in addition to showing our results are not dependent on the continuous-time assumption, will prove useful when deriving comparative statics.

in an interval $[\underline{X}, \bar{X}] \subset (\underline{Y}, \bar{Y})$. Unless stated otherwise, we take $X_0 = Y_0$. The law of motion for X is

$$X_{t+\Delta} = X_t + e_t, \quad e_t \in \{-\epsilon, 0, \epsilon\},$$

where the distribution of e_t is Markovian in X_t and given by $q_+(x) := \mathbb{P}(e_t = \epsilon | X_t = x)$ and $q_-(x) := \mathbb{P}(e_t = -\epsilon | X_t = x)$. We then set $\underline{X} = X_0 - h\epsilon$ and $\bar{X} = X_0 + h\epsilon$, where

$$h = \max \left\{ h' \in \mathbb{N} : (X_0 - h'\epsilon, X_0 + h'\epsilon) \subset (\underline{Y}, \bar{Y}) \right. \\ \left. \text{and } \frac{1}{2} \left(\frac{\sigma^2(x)}{\sigma_0^2} \pm \frac{\mu(x)}{\sigma_0} \sqrt{\Delta} \right) \in (0, 1) \quad \forall x \in (X_0 - h'\epsilon, X_0 + h'\epsilon) \right\}.$$

We take Δ sufficiently small that $\underline{X}, \bar{X}$ are well defined. Whenever discussing a specific value x taken by X_t , we implicitly assume throughout that $x \in \mathcal{G} := \{X_0 \pm k\epsilon : k \in \mathbb{N}, k \leq h\}$. For $x \in (\underline{X}, \bar{X})$, we set

$$q_+(x) = \frac{1}{2} \left(\frac{\sigma^2(x)}{\sigma_0^2} + \frac{\mu(x)}{\sigma_0} \sqrt{\Delta} \right), \quad q_-(x) = \frac{1}{2} \left(\frac{\sigma^2(x)}{\sigma_0^2} - \frac{\mu(x)}{\sigma_0} \sqrt{\Delta} \right),$$

and $q_-(\bar{X}) = 1 > 0 = q_+(\bar{X})$ and $q_-(\underline{X}) = 0 < q_+(\underline{X})$
 $= \min \left\{ 1, \frac{\underline{X}(1 - e^{-r\Delta})}{e^{-r\Delta}\epsilon} \right\}$. Then $e^{-rt}X_t$ is a supermartingale because $rx \geq \mu(x)$ by Assumption 1.

To relate X to Y , we now state a technical Lemma that follows from Theorem 4.1 of Bhattacharya and Waymire (2009, chap. 5) and an application of Skorohod's representation theorem; this Lemma implies that X approximates Y in the limit as $\Delta \rightarrow 0$ (we suppress dependence of X on Δ).

LEMMA A.1: *There exists a probability space on which we can define Y and X such that, for any $T > 0$ and $\delta > 0$, there exists $\bar{\Delta}(\delta, T) > 0$ such that $\mathbb{P}(\max_{t \in [0, T]} |Y_t - X_t| > \delta) < \delta$ for all $\Delta < \bar{\Delta}(\delta, T)$.*

Abusing notation slightly, we redefine $M_t = \max_{s \leq t} X_s$ and let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be the probability space over which X is defined. Without loss, we let $\omega \in \Omega$ be given by $\omega = (x_0, U_0, x_1, U_1, \dots)$, where $X_t(\omega) = x_t$ and $U_t \sim_{iid} U[0, 1]$ is a randomization device to allow for randomized stopping times. We take $\mathcal{F}_t = \sigma(X_0, U_0, \dots, X_t, U_t)$ to be the natural filtration of X, U and let $h_t = (X_0, U_0, \dots, U_{t-\Delta}, X_t)$ denote an arbitrary history up to time t (not including the realization of the period t randomization device U_t).

A contract $(\tau, d_\tau, \alpha_\tau)$ must be such that τ is a stopping time with respect to $\{\mathcal{F}_t\}_{t \geq 0}$ and (d_τ, α_τ) are $\mathcal{F}_\tau$ -measurable. For a fixed sample point $\omega \in \Omega$, $\tau(\omega) \in \{0, \Delta, \dots\}$ is the stopping time induced by ω , with similar definitions for $d_\tau(\omega)$ and $\alpha_\tau(\omega)$. Define the shift operator $\theta_t : \Omega \rightarrow \Omega$ by, for $\omega = (x_0, U^0, \dots, x_t, U_t, x_{t+\Delta}, \dots)$, $\theta_t \omega = (x_t, U_t, x_{t+\Delta}, \dots)$. We say $(\tau', d'_\tau, \alpha'_\tau)$ is the *continuation contract* of $(\tau, d_\tau, \alpha_\tau)$

after history h_t with $\tau(\omega) \geq t$ for all ω generating h_t if $\tau'(\theta_t\omega) = \tau(\omega) - t$, $d'_\tau(\theta_t\omega) = d_\tau(\omega)$, and $\alpha'_\tau(\theta_t\omega) = \alpha_\tau(\omega)$ for all ω generating h_t . A contract $(\tau, d_\tau, \alpha_\tau)$ is *stationary in (X, M)* if, for all histories h_t and h_s such that $(X_t, M_t) = (X_s, M_s) = (x, m)$ and the contract has not stopped, the continuation contracts at h_t and h_s are the same. Finally, abusing notation, we let $\tau_+(x') = \inf\{t: X_t \geq x'\}$ and $\tau_-(x') = \inf\{t: X_t \leq x'\}$ and $J(\tau, d_\tau, \alpha_\tau)$ and $V(\tau, d_\tau, \alpha_\tau)$ be as defined in the main text after replacing Y_τ with X_τ .

We first study a discrete-time version of (1), namely, to find a $(\tau, d_\tau, \alpha_\tau)$ that will maximize $J(\tau, d_\tau, \alpha_\tau)$ subject to *DIR*. Let us provide a quick outline of the analysis of this problem before going into the details. A key difficulty in solving P 's problem is the dynamic individual rationality constraint, as it is difficult to verify whether it holds for any arbitrary contract. To get around this, we relax P 's problem by dropping dynamic individual rationality in place of a weaker set of constraints in which A only considers taking the outside option when $t = \tau_+(m)$ for some m . We choose this relaxation based on a conjecture that the dynamic individual rationality constraint will only bind when the outside option is equal to its maximum. This allows us to apply a Lagrangian approach, with which we can characterize the optimal stopping rule and demand, showing that they are characterized by a demand and generalized version of a split threshold that are measurable in (X, M) . We then analyze how A 's continuation value evolves over time and identify when it is equal to the outside option, which will allow us to verify that the continuous-time limit contract satisfies *DIR*. We conclude by characterizing how the split threshold evolves over time.

Our first lemma simplifies the set of contracts we need to consider.

LEMMA A.2: *There exists a constant $\bar{R}$ and optimal contract $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ such that $\tau^* \leq \tau_+(\bar{R})$ and $d_{\tau^*}^* = \mathbf{1}\{X_{\tau^*} < \bar{R}\}$, and $J(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*) > 0$ if and only if $X_0 < \bar{R}$.*

$\bar{R} \in \mathcal{G}$ is the threshold at which A would take his outside option in the contract that maximizes A 's utility. If P were to let A take his outside option at $X_t < \bar{R}$, then P could do better by promising to make a demand of some sufficiently small $\delta > 0$ in each period and let A decide when to accept it or take his outside option. A would be willing to continue negotiating at X_t and would take a split in the future with strictly positive probability; this change strictly benefits P , who now receives $u_P(\delta)$ with strictly positive probability rather than zero from letting A take his outside option. If $X_0 \geq \bar{R}$, then taking the outside option immediately is optimal. We henceforth restrict attention to optimal contracts $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ with $\tau^* \leq \tau_+(\bar{R})$ and $d_{\tau^*}^* = \mathbf{1}\{X_{\tau^*} < \bar{R}\}$ and assume $X_0 < \bar{R}$.

Next, we relax *DIR* by only considering a subset of “deviations” for A taking his outside option prior to τ . Suppose P chooses contract $(\tau, d_\tau, \alpha_\tau)$ and A plans to take his outside option at $\tau_+(c)$. His expected utility from such a strategy is $V(\tau \wedge \tau_+(c), d_\tau(c), \alpha_\tau)$, where $d_\tau(c) := d_\tau \mathbf{1}\{\tau < \tau_+(c)\}$. We define a relaxed dynamic individual rationality (*RDIR*) constraint for c :

$$RDIR(c): V(\tau \wedge \tau_+(c), d_\tau(c), \alpha_\tau) \leq V(\tau, d_\tau, \alpha_\tau).$$

We relax P 's discrete-time problem by replacing DIR with a set of $RDIR$ constraints.³³ Let $X^n = X_0 + n\epsilon$, and take N such that $X^N = \bar{R} - \epsilon$. We solve a relaxed discrete-time problem (RDP) given by

$$(3) \quad \begin{aligned} &RDP: \sup_{(\tau, d_\tau, \alpha_\tau)} J(\tau, d_\tau, \alpha_\tau) \\ &\text{subject to } RDIR(X^n) \quad \forall n = 0, \dots, N. \end{aligned}$$

Although we can find conditions under which the solution to RDP satisfies DIR , and hence solves the discrete-time version of (1), doing so will be unnecessary for proving Theorem 1.³⁴

For any contract $(\tau, d_\tau, \alpha_\tau)$, we have $\tau \geq \tau_+(X^n)$ if and only if $M_\tau \geq M_{\tau_+(X^n)} = X^n$, so $RDIR(X^n)$ then simplifies to

$$E \left[\left(e^{-r\tau} (d_\tau [u_A(1 - \alpha_\tau) - X_\tau] + X_\tau) - e^{-r\tau_+(X^n)} X^n \right) \mathbf{1}\{M_\tau \geq X^n\} \right] \geq 0.$$

This inequality suggests the use of M_t as a state variable in the relaxed problem. Our next Lemma builds on this observation by using results in Altman (1999) to show properties of a solution to RDP and a Lagrangian method to solve for it. For some multipliers $\Lambda = (\lambda^0, \dots, \lambda^N) \in \mathbb{R}_-^{N+1}$, take the Lagrangian $\mathcal{L}(\Lambda)$ to be

$$(4) \quad \begin{aligned} \mathcal{L}(\Lambda) = \max_{(\tau, d_\tau, \alpha_\tau)} E \left[e^{-r\tau} \left(d_\tau u_P(\alpha_\tau) \right. \right. \\ \left. \left. - \sum_{n=0}^N \lambda^n \mathbf{1}\{M_\tau \geq X^n\} [d_\tau (u_A(1 - \alpha_\tau) - X_\tau) + X_\tau] \right) \right. \\ \left. + \sum_{n=0}^N \lambda^n \mathbf{1}\{M_\tau \geq X^n\} e^{-r\tau_+(X^n)} X^n \right]. \end{aligned}$$

LEMMA A.3: *There exists a solution to RDP that is stationary in (X, M) and solves $\mathcal{L}(\Lambda)$ for some $\Lambda \in \mathbb{R}_-^{N+1}$ such that $\lambda^n < 0$ only if $RDIR(X^n)$ binds.*

We fix a solution $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ to RDP that is stationary in (X, M) and let Λ be the corresponding multipliers. Henceforth, we say “ $(\tau, d_\tau, \alpha_\tau)$ is the continuation contract at (x, m) ” as shorthand for “ $(\tau, d_\tau, \alpha_\tau)$ is the continuation contract of $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ after a history h_t with $(X_t, M_t) = (x, m)$ and $\tau(\omega) \geq t$ for all ω generating h_t .”

Next, we pin down the split amount.

³³That this is a relaxation is implied by Lemma O.A.1 in the online Appendix, which shows any contract that satisfies DIR also satisfies $RDIR(c)$ for any c .

³⁴We will show below that A never has an incentive to take his outside option at $t < \tau$ before observing the outcome of the contract in period t . Because we allow for randomized contracts, the optimal contract could satisfy this property while violating ex post individual rationality (i.e., $u_A(1 - \alpha_t) < X_t$). However, an earlier version of the paper showed that as long as $e^{-rt} Y_t$ is a strict supermartingale and $S^*(X_0) < X_0$, then the solution to RDP satisfies DIR for sufficiently small Δ .

LEMMA A.4: *There exists a decreasing function $\alpha(m)$ such that $\alpha_{\tau^*}^* = \alpha(M_{\tau^*})$.*

PROOF:

Pointwise optimization of $\alpha_{\tau^*}^*$ in (4) implies that

$$\alpha_{\tau^*}^* = \alpha(M_{\tau^*}) := \arg \max_{\alpha \in [0,1]} u_P(\alpha) - \sum_{n=0}^N \lambda^n \mathbf{1}\{M_{\tau^*} \geq X^n\} u_A(1 - \alpha).$$

Strict concavity of u_P and u_A implies this argmax is unique. Because $\lambda^n \leq 0$, $-\sum_{n=0}^N \lambda^n \mathbf{1}\{m \geq X^n\}$ is increasing in m , so $u_P(\alpha) - \sum_{n=0}^N \lambda^n \mathbf{1}\{m \geq X^n\} u_A(1 - \alpha)$ is submodular in α and m . Topkis's Theorem then implies $\alpha(m)$ is decreasing. ■

We now study the optimal stopping rule τ^* . We will show that, as long as M_t is constant, the stopping rule will (essentially) take a threshold form, stopping whenever X_t goes below some threshold, potentially subject to some randomization on whether to stop when exactly at the threshold (as we will see, the threshold and mixing probability may change as M_t changes). We begin by looking at when stopping is optimal at $t < \tau_+(X^1)$ (i.e., when $M_t = X_0$). We say a stopping time uses an (S, γ) -split threshold if it stops with probability $\gamma \in [0, 1)$ in each period t with $X_t = S + \epsilon$ and with probability one when $X_t \leq S$; we denote such stopping times by $\tau^{S, \gamma}$.³⁵ If the optimal contract stops immediately (i.e., $\mathbb{P}(\tau^* = 0) = 1$), then τ^* is trivially an (S, γ) -split threshold for $S = X_0$. Our next lemma examines the case when the contract does not stop immediately but does stop prior to $\tau_+(X^1)$ with positive probability and shows τ^* uses an (S, γ) -split threshold prior to $\tau_+(X^1)$; for the case when $\mathbb{P}(\tau^* < \tau_+(X^1)) = 0$, the same result trivially holds, taking $S < \underline{X}$ and $\gamma = 0$. The S^0, γ^0 we identify below will correspond to the stopping threshold used when $M_t = X_0$.

LEMMA A.5: *If $\mathbb{P}(\tau^* \in (0, \tau_+(X^1))) > 0$, then there is a unique (S^0, γ^0) such that $\tau^* = \tau^{S^0, \gamma^0}$ when $\tau^* < \tau_+(X^1)$.*

PROOF:

Suppose $\mathbb{P}(\tau^* \in (0, \tau_+(X^1))) > 0$. Because $M_{\tau^*} \geq X^1$ if and only if $\tau^* \geq \tau_+(X^1)$ (in which case $d_{\tau^*}^* = 1$), the payoff in $\mathcal{L}(\Lambda)$ (under the optimal contract) from stopping at $\tau^* < \tau_+(X^1)$ is

$$\begin{aligned} d_{\tau^*}^* u_P(\alpha_{\tau^*}^*) - \sum_{n=0}^N \lambda^n \mathbf{1}\{M_{\tau^*} \geq X^n\} [d_{\tau^*}^* (u_A(1 - \alpha_{\tau^*}^*) - X_{\tau^*}) + X_{\tau^*}] \\ = u_P(\alpha(X_0)) - \lambda^0 u_A(1 - \alpha(X_0)). \end{aligned}$$

³⁵There are many ways to formally define $\tau^{S, \gamma}$ using the randomization device $\{U_t\}_{t=0}^\infty$ (e.g., $\tau^{S, \gamma} = \inf\{t: X_t \leq S, \text{ or } X_t = S + \epsilon \text{ and } U_t \leq \gamma\}$) that differ only in which realizations of U_t will activate the stopping time. We will treat all such stopping times identically, as the exact specification of which values of U_t lead to stopping is immaterial for our results.

Let $\bar{u}(\lambda^0) := u_P(\alpha(X_0)) - \lambda^0 u_A(1 - \alpha(X_0)) > 0$. Let $K(X^1)$ be the continuation value in $\mathcal{L}(\Lambda)$ at $\tau_+(X^1)$ and $L(x, \tau)$ be the continuation value at $(X_t, M_t) = (x, X_0)$ from using $(\tau, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ until $\tau_+(X^1)$ and reverting to the optimal continuation strategy at $\tau_+(X^1)$:

$$L(x, \tau) := E_x \left[e^{-r\tau} \bar{u}(\lambda^0) \mathbf{1}\{\tau < \tau_+(X^1)\} + e^{-r\tau_+(X^1)} K(X^1) \mathbf{1}\{\tau \geq \tau_+(X^1)\} \right].$$

Let $L^*(x) = \max_{\tau} L(x, \tau)$. Then $L^*(X_0) = L(X_0, \tau^*)$,³⁶ and $L^*(x) \geq \bar{u}(\lambda^0)$ for all $x < X^1$. Let $\mathcal{C}$ be the continuation region defined by

$$\mathcal{C} := \{x : x < X^1, \exists \tau \text{ s.t. } L(x, \tau) \geq \bar{u}(\lambda^0) \text{ and } \mathbb{P}(\tau > 0) > 0\}.$$

By standard optimal stopping arguments (Peskir and Shiryaev 2006), continuing at x is optimal (at least weakly) if $x \in \mathcal{C}$ and stopping at x is strictly optimal if $x \notin \mathcal{C}$. That $\mathbb{P}(\tau^* > 0) > 0$ implies $X_0 \in \mathcal{C}$.

Let $S' = \max\{x : x < X^1, L^*(x) = \bar{u}(\lambda^0)\}$. Such an S' exists because $\mathbb{P}(\tau^* < \tau_+(X^1)) > 0$ and, by standard optimal stopping arguments, stopping at $t < \tau_+(X^1)$ is optimal if and only if $L^*(X_t) = \bar{u}(\lambda^0)$; thus, it is never optimal to stop at $t < \tau_+(X^1)$ with $X_t > S'$. We now argue it is strictly optimal to stop immediately at all $x < S'$. For the sake of contradiction, suppose there exists $x < S'$ such that $x \in \mathcal{C}$. Let $\tau' = \inf\{t : X_t \notin \mathcal{C} \text{ or } X_t = S'\}$, so $L^*(X_{\tau'}) = \bar{u}(\lambda^0)$ if $\tau' < \tau_+(X^1)$. Note that $\mathbb{P}_x(\tau' < \tau_+(X^1)) = 1$ by $x < S'$. Standard dynamic programming arguments then imply

$$L^*(x) = E_x[e^{-r\tau'} L^*(X_{\tau'})] = E_x[e^{-r\tau'} \bar{u}(\lambda^0)] < \bar{u}(\lambda^0),$$

a contradiction of $L^*(x) \geq \bar{u}(\lambda^0)$. Thus, $x \notin \mathcal{C}$ for all $x < S'$; namely, stopping is strictly optimal at all $x < S'$.

Recall that for all $t < \tau_+(X^1)$, M_t is constant (and equal to X_0). For any stopping time that is stationary in (X, M) , the probability of stopping at a date $t < \tau_+(X^1)$ (conditional on not having previously stopped) can only depend on X_t because M_t is constant across such t . Therefore, every solution to $\mathcal{L}(\Lambda)$ that is stationary in (X, M) must either stop with probability one when $X_t \leq S'$ and $t < \tau_+(X^1)$, which is an $(S', 0)$ -split threshold (in which case $S^0 = S', \gamma^0 = 0$), or must stop with some constant probability³⁷ $\gamma \in [0, 1)$ whenever $X_t = S'$ at $t < \tau_+(X^1)$ and stop with probability one at all t with $X_t \leq S' - \epsilon$, which is an $(S' - \epsilon, \gamma)$ -split threshold (in which case $S^0 = S' - \epsilon, \gamma^0 = \gamma$). ■

³⁶If there was a solution τ' to $L^*(x)$ that generated a strictly higher value than τ^* , we could construct a stopping time employing τ' prior to $\tau_+(X^1)$ and τ^* after $\tau_+(X^1)$ that generates a strictly higher value in the Lagrangian than τ^* , a contradiction of the fact that τ^* solves the Lagrangian.

³⁷It is a constant probability because the decision to stop is stationary in (X, M) .

The above results give the optimal contract for t such that $M_t = X_0$. We now turn to the optimal continuation contract when M_t increases to X^1 (i.e., at $\tau_+(X^1)$). The continuation contract at $\tau_+(X^1)$ solves

$$\begin{aligned} K(X^1) = \max_{(\tau, d_\tau, \alpha_\tau)} E_{X^1} & \left[e^{-r\tau} \left(d_\tau u_P(\alpha_\tau) \right. \right. \\ & \left. \left. - \sum_{n=0}^N \lambda^n \mathbf{1}\{M_\tau \geq X^n\} [d_\tau(u_A(1 - \alpha_\tau) - X_\tau) - X_\tau] \right) \right. \\ & \left. + \sum_{n=1}^N \lambda^n \mathbf{1}\{M_\tau \geq X^n\} e^{-r\tau_+(X^n)} X^n \right]. \end{aligned}$$

We can apply the same arguments as in Lemma A.5 when solving $K(X^1)$ to conclude that, for some (S^1, γ^1) , an (S^1, γ^1) -split threshold is used between $\tau_+(X^1)$ and $\tau_+(X^2)$ (i.e., when $M_t = X^1$). Repeating these arguments, we can show that, for each n , there exists (S^n, γ^n) such that the optimal continuation contract at t uses an (S^n, γ^n) -split threshold at $t \in [\tau_+(X^n), \tau_+(X^{n+1})]$. Because $M_t = X^n$ if and only if $t \in [\tau_+(X^n), \tau_+(X^{n+1})]$, we can write the prevailing (S, γ) as a function of M_t . Define $(S(\cdot), \gamma(\cdot))$ by $S(X^n) := S^n$ and $\gamma(X^n) := \gamma^n$. Without loss, we take $S(m) \leq m$.³⁸

By Lemma A.3, $\lambda^n = 0$ whenever $RDIR(X^n)$ is slack, in which case the continuation value in $\mathcal{L}(\Lambda)$ is the same at $(X_t, M_t) = (x, X^n)$ and $(X_t, M_t) = (x, X^n - \epsilon)$ whenever $RDIR(X^n)$ is slack. Intuitively, this suggests that the contract should be stationary in m across regions where the $RDIR$ constraints are slack. We show this formally in Lemma O.A.2 in the online Appendix, namely, that there is an optimal contract that is stationary in (X, M) such that, for any $X^k < X^\ell$ with $RDIR(X^n)$ slack for all $X^n \in (X^k, X^\ell)$, the corresponding $S(m), \alpha(m), \gamma(m)$ are constant over $[X^k, X^\ell]$. Without loss, we assume $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ satisfies this property.

We now derive additional structure on the optimal contract and A 's continuation value. Our next lemma proves a useful bound on $S(m)$, which implies that it is never optimal to take a split with probability one if $S(X_0) < X_0$.

LEMMA A.6: *If $S(X_0) < X_0$, then $S(m) < m$ for all $m \in (X_0, \bar{R})$.*

PROOF:

For the sake of contradiction, suppose $S(X_0) < X_0$ and $S(m) = m$ for some $m \in (X_0, \bar{R})$. Because $S(m) = m$, $\mathbb{P}(\tau^* \leq \tau_+(m)) = 1$, which implies $\mathbb{P}(d_{\tau^*}^* = 0) = 0$. Letting $\bar{\alpha} = E[\alpha_{\tau^*}^*]$, we have

$$V(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*) = E[e^{-r\tau^*} u_A(1 - \alpha_{\tau^*}^*)] \leq E[u_A(1 - \alpha_{\tau^*}^*)] \leq u_A(1 - \bar{\alpha}),$$

³⁸ Setting $S(m)$ to be $S' \geq m$ implies stopping immediately at $\tau_+(m)$, so all such S' are equivalent.

where the first inequality is strict whenever $\mathbb{P}(\tau^* > 0, \alpha_{\tau^*}^* < 1) > 0$ and the second inequality follows from Jensen's inequality. Similarly, $J(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*) \leq u_P(\bar{\alpha})$, strictly so if $\mathbb{P}(\tau^* > 0, \alpha_{\tau^*}^* > 0) > 0$.

$S(X_0) < X_0$ implies $\mathbb{P}(\tau^* > 0) > 0$, so either $\mathbb{P}(\tau^* > 0, \alpha_{\tau^*}^* < 1) > 0$ or $\mathbb{P}(\tau^* > 0, \alpha_{\tau^*}^* > 0) > 0$. Thus, either $J(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*) < u_P(\bar{\alpha})$ or $V(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*) < u_A(1 - \bar{\alpha})$. P can do strictly better than $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ by implementing an immediate split, in the former case, of $\bar{\alpha}$,³⁹ and, in the latter case, of $\bar{\alpha} + \delta$ for some $\delta > 0$ such that $u_A(1 - \bar{\alpha} - \delta) > X_0$, a contradiction of the optimality of $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$. ■

Using stationarity of $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ in (X, M) , we now simplify the *RDIR* constraint. Let $\mathcal{R}$ be the set of (x, m) that are reached with positive probability prior to τ^* :

$$\mathcal{R} = \{(x, m) : \mathbb{P}(\exists t \leq \tau^* \text{ s.t. } (X_t, M_t) = (x, m)) > 0\}.$$

For $(x, m) \in \mathcal{R}$, with some abuse of notation, we let $J(x, m)$ and $V(x, m)$ be P 's and A 's respective continuation values from $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ at the beginning of a period t with $(X_t, M_t) = (x, m)$ prior to the realization of U_t .⁴⁰ *RDIR*(m) is then equivalent to

$$\begin{aligned} & E\left[e^{-r\tau^*} u_A(1 - \alpha_{\tau^*}^*) \mathbf{1}\{\tau^* < \tau_+(m)\} + e^{-r\tau_+(m)} m \mathbf{1}\{\tau^* \geq \tau_+(m)\}\right] \\ & \leq E\left[e^{-r\tau^*} u_A(1 - \alpha_{\tau^*}^*) \mathbf{1}\{\tau^* < \tau_+(m)\} + e^{-r\tau_+(m)} V(m, m) \mathbf{1}\{\tau^* \geq \tau_+(m)\}\right]. \end{aligned}$$

For $(m, m) \in \mathcal{R}$, *RDIR*(m) then simplifies to $m \leq V(m, m)$, with equality if and only if *RDIR*(m) binds.

The next lemma provides bounds on A 's continuation value that will allow us to verify that the continuous-time limit of the solution to the relaxed problem satisfies *DIR*.

LEMMA A.7: $V(x, m) \geq x$ for all $(x, m) \in \mathcal{R}$, with equality if $x = m$.

PROOF:

The proof is immediate if $S(X_0) = X_0$, as in this case $\mathcal{R} = \{(X_0, X_0)\}$ and *RDIR*(X_0) must bind, as otherwise, P could increase his demands slightly while still satisfying all *RDIR* constraints.

Suppose $S(X_0) < X_0$. Then, $S(m) < m$ for all $m \in [X_0, \bar{R}]$ by Lemma A.6, which implies $(m, m) \in \mathcal{R}$. Because the outside option is taken at $\bar{R}$, we have $\{(x, \bar{R}) \in \mathcal{R}\} = \{(\bar{R}, \bar{R})\}$ and $V(\bar{R}, \bar{R}) = \bar{R}$.

We proceed to prove the lemma by induction, showing $V(x, m) \geq x$ and $V(m, m) = m$ whenever $V(m + \epsilon, m + \epsilon) = m + \epsilon$. Take an $m \in [X_0, \bar{R}]$ such that $V(m + \epsilon, m + \epsilon) = m + \epsilon$; these properties hold for $m = \bar{R} - \epsilon$. Suppose there exists $x < m$ such that $(x, m) \in \mathcal{R}$ and $V(x, m) \leq x$. Let $(\tau', d_{\tau'}', \alpha_{\tau'}')$ be

³⁹Such an immediate split satisfies *DIR* as $V(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*) \leq u_A(1 - \bar{\alpha})$ and, by *RDIR*(X_0), $V(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*) \geq V(\tau^* \wedge \tau_+(X_0), d_{\tau^*}^*(X_0), \alpha_{\tau^*}^*) = X_0$.

⁴⁰That is, if the contract involves randomization at period t , we take the expected utility of each player prior to the randomization taking place.

the continuation contract at $\tau_+(m)$. Because $(x, m) \in \mathcal{R}$, $x \geq S(m)$.⁴¹ Note that $\mathbb{P}_m(\tau' < \tau_+(m + \epsilon) \wedge \tau_-(x)) > 0$ if and only if $x = S(m)$ and $\gamma(m) > 0$. In this case, conditional on $\tau' < \tau_+(m + \epsilon) \wedge \tau_-(x)$, because $V(x, m) \leq x$ by assumption and $x = S(m)$, we have $X_{\tau'} = S(m) + \epsilon > S(m) \geq V(S(m), m) = u_A(1 - \alpha(m))$. We then have

$$\begin{aligned}
 (5) \quad V(m, m) &= E_m \left[e^{-r\tau'} u_A(1 - \alpha(m)) \mathbf{1}\{\tau' < \tau_+(m + \epsilon) \wedge \tau_-(x)\} \right. \\
 &\quad + e^{-r\tau_-(x)} V(x, m) \mathbf{1}\{\tau_-(x) \leq \tau' \wedge \tau_+(m + \epsilon)\} \\
 &\quad \left. + e^{-r\tau_+(m+\epsilon)} V(m + \epsilon, m + \epsilon) \mathbf{1}\{\tau_+(m + \epsilon) \leq \tau' \wedge \tau_-(x)\} \right] \\
 &\leq E_m \left[e^{-r\tau'} X_{\tau'} \mathbf{1}\{\tau' < \tau_+(m + \epsilon) \wedge \tau_-(x)\} \right. \\
 &\quad + e^{-r\tau_-(x)} x \mathbf{1}\{\tau_-(x) \leq \tau' \wedge \tau_+(m + \epsilon)\} \\
 &\quad \left. + e^{-r\tau_+(m+\epsilon)} (m + \epsilon) \mathbf{1}\{\tau_+(m + \epsilon) \leq \tau' \wedge \tau_-(x)\} \right] \\
 &= E_m \left[e^{-r[\tau' \wedge \tau_+(m+\epsilon) \wedge \tau_-(x)]} X_{\tau' \wedge \tau_+(m+\epsilon) \wedge \tau_-(x)} \right] \\
 &\leq m,
 \end{aligned}$$

where the last line follows from Doob's optional-stopping theorem. Because $x \geq S(m)$, $\mathbb{P}_m(\tau_-(x) \leq \tau' \wedge \tau_+(m + \epsilon)) > 0$. Thus, the first inequality in (5) is strict if $V(x, m) < x$, which contradicts $V(m, m) \geq m$ by $RDIR(m)$. Therefore, $V(x, m) \geq x$. Moreover, (5) implies that if $V(m, m) > m$, then $V(x, m) > x$ for all $(x, m) \in \mathcal{R}$.

We conclude by showing $V(m, m) = m$. Suppose $V(m, m) > m$; then $RDIR(m)$ is slack. $RDIR(X_0)$ must bind,⁴² so $m > X_0$. Let $\hat{m} = \max\{m' : m' < m, RDIR(m') \text{ binds}\}$. Because no $RDIR$ constraints bind between $\hat{m}$ and m , $\alpha(\cdot)$, $S(\cdot)$, $\gamma(\cdot)$ are all constant over $[\hat{m}, m]$, which implies $V(\hat{m}, \cdot)$ is constant over $[\hat{m}, m]$. Thus, $V(\hat{m}, m) = V(\hat{m}, \hat{m})$, and, by $RDIR(\hat{m})$ binding, $V(\hat{m}, \hat{m}) = \hat{m}$. But the observation following (5) when $x = \hat{m}$ and $V(m, m) > m$ implies $V(\hat{m}, m) > \hat{m}$, a contradiction. Thus, $V(m, m) = m$. ■

We now turn to showing some properties of α and P 's continuation value.

LEMMA A.8: $J(m, m) \leq u_P(\alpha(m))$, $u_A(1 - \alpha(m)) \geq S(m)$, and $\alpha(m) > 0$ for all $m \in [X_0, \bar{R})$.

PROOF:

Take $m \in [X_0, \bar{R})$. Because M_t only increases over time, we have

$$(6) \quad J(m, m) = E_m \left[e^{-r\tau^*} d_{\tau^*}^* u_P(\alpha(M_{\tau^*})) \right] \leq E_m \left[e^{-r\tau^*} d_{\tau^*}^* u_P(\alpha(m)) \right] \leq u_P(\alpha(m)).$$

⁴¹ Otherwise, τ^* would stop at probability one before (x, m) is reached, contradicting $(x, m) \in \mathcal{R}$.

⁴² If it were slack, P could implement a split with demand $\alpha = 1$ with some small probability at $t = 0$ and strictly increase his own utility.

Because $V(S(m), m) = u_A(1 - \alpha(m))$ and, by Lemma A.7, $V(S(m), m) \geq S(m)$, we have $u_A(1 - \alpha(m)) \geq S(m)$.

Suppose $\alpha(m) = 0$. Then (6) implies $J(m, m) = 0$. By Lemma A.7, $V(m, m) = m$, so both players receive the same continuation value at $\tau_+(m)$ as if they took the outside option. But Lemma A.2 implies that there exists a continuation contract at $\tau_+(m)$ that satisfies *DIR* and strictly increases P 's continuation value, contradicting the optimality of $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$. Thus, $\alpha(m) > 0$. ■

We conclude the analysis of *RDP* by looking at the dynamics of $S(\cdot)$. We focus on the case of $S(X_0) < X_0$, as the structure of $S(\cdot)$ above X_0 has no impact of outcomes if $S(X_0) = X_0$; in such a case, we take $S(\cdot)$ decreasing without loss.

LEMMA A.9: $S(\cdot)$ is decreasing on $[X_0, \bar{R})$ if $S(X_0) < X_0$.

PROOF:

Suppose $S(X_0) < X_0$ and take $m \in [X_0, \bar{R})$. Let $(S_1, \gamma_1) = (S(m), \gamma(m))$ and $(S_2, \gamma_2) = (S(m + \epsilon), \gamma(m + \epsilon))$. For the sake of contradiction, suppose $S_1 < S_2$. Define, for $i \in \{1, 2\}$,

$$\begin{aligned}\Phi_i &= E_m \left[e^{-r\tau_+(m+\epsilon)} \mathbf{1} \{ \tau_+(m + \epsilon) \leq \tau^{S_i, \gamma_i} \} \right], \\ \phi_i &= E_m \left[e^{-r\tau^{S_i, \gamma_i}} \mathbf{1} \{ \tau^{S_i, \gamma_i} < \tau_+(m + \epsilon) \} \right].\end{aligned}$$

Starting at $X_0 = m$, we consider an alternative problem in which we fix continuation values for P and A at $\tau_+(m + \epsilon)$ to be $J(m + \epsilon, m + \epsilon)$ and $m + \epsilon$, respectively, and allow P to choose a fixed split amount and an (S, γ) -split threshold in $\{(S_1, \gamma_1), (S_2, \gamma_2)\}$ subject to, for some $W > 0$, a promise-keeping constraint to deliver a continuation value at $t = 0$ of W to A . The promise-keeping constraint below uniquely pins down the split amount for each choice of (S_i, γ_i) , which we denote by $\alpha_i(W)$:

$$(7) \quad W = \Phi_i \cdot (m + \epsilon) + \phi_i \cdot u_A(1 - \alpha_i(W)).$$

$\alpha_i(W)$ is clearly decreasing in W . P 's expected utility in this new problem is

$$(8) \quad \hat{J}(W) := \max_{i \in \{1, 2\}} \Phi_i \cdot J(m + \epsilon, m + \epsilon) + \phi_i \cdot u_P(\alpha_i(W)).$$

$\hat{J}(W)$ is decreasing in W because each $\alpha_i(W)$ is decreasing in W .

We now show $\hat{J}(V(m, m)) = J(m, m)$ and $i = 1$ is an optimal choice in (8) when $W = V(m, m)$. Note that $\alpha(m) = \alpha_1(V(m, m))$ because

$$\begin{aligned}V(m, m) &= E_m \left[e^{-r\tau^{S(m), \gamma(m)}} u_A(1 - \alpha(m)) \mathbf{1} \{ \tau^{S(m), \gamma(m)} < \tau_+(m + \epsilon) \} \right. \\ &\quad \left. + e^{-r\tau_+(m+\epsilon)} V(m + \epsilon, m + \epsilon) \mathbf{1} \{ \tau^{S(m), \gamma(m)} \geq \tau_+(m + \epsilon) \} \right] \\ &= \Phi_1 \cdot (m + \epsilon) + \phi_1 \cdot u_A(1 - \alpha(m)).\end{aligned}$$

Similarly, $J(m, m)$ can be written as $J(m, m) = \Phi_1 \cdot J(m + \epsilon, m + \epsilon) + \phi_1 \cdot u_P(\alpha(m))$. If $i = 1$ was not a solution to $\hat{J}(V(m, m))$, then $J(m, m) < \hat{J}(V(m, m))$. P could then strictly increase his continuation value in RDP at (m, m) , while keeping A 's continuation value at $\tau_+(m)$ the same by replacing the continuation contract at (m, m) with one using a (S_2, γ_2) -split threshold for $t \in [\tau_+(m), \tau_+(m + \epsilon))$ and a demand of $\alpha_2(V(m, m))$ before reverting back at $\tau_+(m + \epsilon)$ to the original continuation contract at $(m + \epsilon, m + \epsilon)$, contradicting the optimality of our original contract in RDP . An analogous argument implies $\hat{J}(V(m, m + \epsilon)) = J(m, m + \epsilon)$ and $i = 2$ is an optimal choice in (8) when $W = V(m, m + \epsilon)$.

For an arbitrary W , $i = 1$ is an optimal choice if and only if

$$(9) \quad \Phi_1 \cdot J(m + \epsilon, m + \epsilon) + \phi_1 \cdot u_P(\alpha_1(W)) - [\Phi_2 \cdot J(m + \epsilon, m + \epsilon) + \phi_2 \cdot u_P(\alpha_2(W))] \geq 0.$$

When we increase W , (9) changes by

$$(10) \quad \phi_1 \cdot u'_P(\alpha_1(W)) \alpha'_1(W) - \phi_2 \cdot u'_P(\alpha_2(W)) \alpha'_2(W).$$

Optimality of $i = 1$ when $W = V(m, m)$ implies (9) holds at $W = V(m, m)$. Similarly, optimality of $i = 2$ when $W = V(m, m + \epsilon)$ implies the reverse inequality in (9) holds at $W = V(m, m + \epsilon)$. Therefore, there must exist a $W \in [V(m, m), V(m, m + \epsilon)]$ at which (9) holds with equality and (10) is negative.⁴³ Fix such a W . Using (7), we have $\alpha'_i(W) = \frac{-1}{\phi_i \cdot u'_A(1 - \alpha_i(W))}$, and so (10) being negative simplifies to

$$(11) \quad \frac{u'_P(\alpha_2(W))}{u'_A(1 - \alpha_2(W))} \leq \frac{u'_P(\alpha_1(W))}{u'_A(1 - \alpha_1(W))}.$$

By strict concavity of either u_P or u_A , (11) implies $\alpha_1(W) \leq \alpha_2(W)$.

Note that $S_1 < S_2$ implies $\mathbb{P}_m(\tau^{S_1, \gamma_1} \geq \tau^{S_2, \gamma_2}) = 1$ and $\mathbb{P}_m(\tau^{S_1, \gamma_1} > \tau^{S_2, \gamma_2}) > 0$; moreover, $\mathbb{P}_m(\tau^{S_1, \gamma_1} = \tau^{S_2, \gamma_2}) > 0$ if and only if $S_1 = S_2 - \epsilon$ and $\gamma_1 > 0$. Under (S_1, γ_1) , P 's continuation value at τ^{S_2, γ_2} conditional on $\tau^{S_2, \gamma_2} < \tau_+(m + \epsilon) \wedge \tau^{S_1, \gamma_1}$ and $X_{\tau^{S_2, \gamma_2}} = x > S^1$ is⁴⁴

$$(12) \quad \tilde{J}(x) := E_x \left[e^{-r\tau^{S_1, \gamma_1}} u_P(\alpha_1(W)) \mathbf{1}\{\tau^{S_1, \gamma_1} < \tau_+(m + \epsilon)\} + e^{-r\tau_+(m + \epsilon)} J(m + \epsilon, m + \epsilon) \mathbf{1}\{\tau^{S_1, \gamma_1} \geq \tau_+(m + \epsilon)\} | \tau^{S_1, \gamma_1} > 0 \right].$$

⁴³Note that, if $V(m, m + \epsilon) = V(m, m)$, then S_1 cannot be optimal, as P could deliver the same continuation value to A by using (S_2, γ_2) and achieve a split sooner, contradicting the optimality of (S_1, γ_1) . Therefore, $V(m, m) < V(m, m + \epsilon)$.

⁴⁴We condition on $\tau^{S_1, \gamma_1} > 0$ in (12) because we are normalizing time to start at 0 in the expectation rather than τ^{S_2, γ_2} and the distribution of $(\tau^{S_1, \gamma_1} - \tau^{S_2, \gamma_2}, X_{\tau^{S_1, \gamma_1}}, \tau_+(m + \epsilon) - \tau^{S_2, \gamma_2})$ conditional on $\tau^{S_2, \gamma_2} < \tau_+(m + \epsilon) \wedge \tau^{S_1, \gamma_1}$, $X_0 = m$, and $X_{\tau^{S_2, \gamma_2}} = x$ is equal to the distribution of $(\tau^{S_1, \gamma_1}, X_{\tau^{S_1, \gamma_1}}, \tau_+(m + \epsilon))$ conditional on $\tau^{S_1, \gamma_1} > 0$ and $X_0 = x$.

Because $J(m + \epsilon, m + \epsilon) \leq u_P(\alpha(m + \epsilon)) = u_P(\alpha_2(V(m + \epsilon, m))) \leq u_P(\alpha_2(W))$ and $u_P(\alpha_1(W)) \leq u_P(\alpha_2(W))$, (12) implies $\tilde{J}(X_{\tau^{S_2, \gamma_2}}) < u_P(\alpha_2(W))$.

P 's expected utility from using (S_1, γ_1) is

$$(13) \quad E_m \left[e^{-r\tau^{S_2, \gamma_2}} u_P(\alpha_1(W)) \mathbf{1}\{\tau^{S_1, \gamma_1} = \tau^{S_2, \gamma_2} < \tau_+(m + \epsilon)\} \right. \\ \left. + e^{-r\tau^{S_2, \gamma_2}} \tilde{J}(X_{\tau^{S_2, \gamma_2}}) \mathbf{1}\{\tau^{S_2, \gamma_2} < \tau_+(m + \epsilon) \wedge \tau^{S_1, \gamma_1}\} \right. \\ \left. + e^{-r\tau_+(m + \epsilon)} J(m + \epsilon, m + \epsilon) \mathbf{1}\{\tau^{S_2, \gamma_2} \geq \tau_+(m + \epsilon)\} \right],$$

whereas P 's expected utility from using (S_2, γ_2) is

$$(14) \quad E_m \left[e^{-r\tau^{S_2, \gamma_2}} u_P(\alpha_2(W)) \mathbf{1}\{\tau^{S_2, \gamma_2} < \tau_+(m + \epsilon)\} \right. \\ \left. + e^{-r\tau_+(m + \epsilon)} J(m + \epsilon, m + \epsilon) \mathbf{1}\{\tau^{S_2, \gamma_2} \geq \tau_+(m + \epsilon)\} \right].$$

By assumption that both $i = 1, 2$ are optimal at W , (13) and (14) are equal. Setting them equal to each other and simplifying, we get

$$0 = E_m \left[e^{-r\tau^{S_2, \gamma_2}} [u_P(\alpha_1(W)) - u_P(\alpha_2(W))] \mathbf{1}\{\tau^{S_2, \gamma_2} = \tau^{S_1, \gamma_1} < \tau_+(m + \epsilon)\} \right. \\ \left. + e^{-r\tau^{S_2, \gamma_2}} [\tilde{J}(X_{\tau^{S_2, \gamma_2}}) - u_P(\alpha_2(W))] \mathbf{1}\{\tau^{S_2, \gamma_2} < \tau^{S_1, \gamma_1} \wedge \tau_+(m + \epsilon)\} \right].$$

The right-hand side is strictly negative because $u_P(\alpha_1(W)) \leq u_P(\alpha_2(W))$, $\tilde{J}(X_{\tau^{S_2, \gamma_2}}) < u_P(\alpha_2(W))$, and $\mathbb{P}_m(\tau^{S_2, \gamma_2} < \tau^{S_1, \gamma_1} \wedge \tau_+(m + \epsilon)) > 0$, a contradiction. Therefore, $S_1 \geq S_2$. ■

Having characterized a solution to RDP , we take the limit as $\Delta \rightarrow 0$ to define our continuous-time contract. We now treat time as continuous, in which case, for any $t \notin \{0, \Delta, 2\Delta, \dots\}$, we define X_t equal to X_s for $s = \max\{s': s' \in \{0, \Delta, \dots\}, s' < t\}$; we keep dependence of X on Δ implicit. Let $(\tau^\Delta, d_{\tau^\Delta}^\Delta, \alpha_{\tau^\Delta}^\Delta)$ be our solution to RDP when the period length is Δ , and let $(S^\Delta(\cdot), \alpha^\Delta(\cdot), \bar{R}^\Delta)$ be the corresponding functions and constant we have derived. Let $(S^*, \alpha^*, \bar{R}^*) = \lim_{\Delta \rightarrow 0} (S^\Delta, \alpha^\Delta, \bar{R}^\Delta)$.⁴⁵ We use $(S^*, \alpha^*, \bar{R}^*)$ to define the contract $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ as in Theorem 1. If $S^*(Y_0) \geq Y_0$, we take $\tau^* = 0, d_{\tau^*}^* = \mathbf{1}\{Y_0 < \bar{R}^*\}$ and $\alpha_{\tau^*}^* = 1 - u_A^{-1}(Y_0)$. That S^* and α^* are decreasing follows from the fact that S^Δ and α^Δ are decreasing.

We now complete the Proof of Theorem 1; we assume below Y and X are defined on a probability space as in Lemma A.1.

PROOF OF THEOREM 1:

That P and A 's expected discrete-time utilities from $(\tau^\Delta, d_{\tau^\Delta}^\Delta, \alpha_{\tau^\Delta}^\Delta)$ converge to their expected continuous-time utility from $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ follows from Lemma A.1. This and the fact that, by Lemma A.7, A 's continuation value $V(X_t, M_t)$ in RDP under $(\tau^\Delta, d_{\tau^\Delta}^\Delta, \alpha_{\tau^\Delta}^\Delta)$ is above the outside option after each history imply $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ satisfies DIR .

⁴⁵ Because S^Δ and α^Δ are decreasing and uniformly bounded, by Helly's Selection Theorem, the sequence as $\Delta \rightarrow 0$ has a subsequence converging to some decreasing functions S^* and α^* .

We next show that $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ is optimal in continuous time. For the sake of contradiction, suppose there exists a *DIR* continuous-time contract $(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**})$ such that $J(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**}) > J(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ (where $J(\tau, d_{\tau}, \alpha_{\tau})$ again denotes the continuous-time utility from $(\tau, d_{\tau}, \alpha_{\tau})$).

To prove such a contract cannot exist, we first expand the action space for contracts in a way that plays no role in the optimal contract. We start by extending u_P and u_A to be defined on $\mathbb{R}$ rather than $[0, 1]$ in any way that satisfies $u_i'' \leq 0 < u_i'$ for $i = P, A$, with each u_i' uniformly bounded away from zero and ∞ .⁴⁶ We also expand the range of α_{τ} to $(-\infty, 1]$. With this extended environment, all the steps in the derivation of the optimal discrete-time contract go through verbatim, and the optimal contract takes the same form as when we restricted to $\alpha_{\tau} \in [0, 1]$.⁴⁷

Take some $\delta_{\alpha} > 0$ and T . If $(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**})$ satisfies *DIR*, then $(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**} - \delta_{\alpha})$ does as well. By the construction in Theorem 8.2.4 of Kushner (1977), there exists a sequence of contracts $(\hat{\tau}^{\Delta}, \hat{d}_{\hat{\tau}^{\Delta}}^{\Delta}, \hat{\alpha}_{\hat{\tau}^{\Delta}}^{\Delta})$ in the discrete-time setting, indexed by the period length Δ , that converges in probability to $(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**} - \delta_{\alpha})$ on $[0, T]$ as $\Delta \rightarrow 0$. If $\hat{\tau}^{\Delta}$ has not stopped by T , we set it to stop immediately at $\min\{t \in \{0, \Delta, \dots\} : t > T\}$ and implement a split with $\hat{\alpha}_{\hat{\tau}^{\Delta}}^{\Delta} < 1 - u_A^{-1}(\bar{Y}) - \delta_{\alpha}$. Because $\delta_{\alpha} > 0$ and $(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**} - \delta_{\alpha})$ satisfies *DIR*, it is easy to see $(\hat{\tau}^{\Delta}, \hat{d}_{\hat{\tau}^{\Delta}}^{\Delta}, \hat{\alpha}_{\hat{\tau}^{\Delta}}^{\Delta})$ satisfies all *RDIR* constraints in the discrete-time setting for sufficiently small Δ . Taking $\delta_{\alpha} \rightarrow 0$ and $T \rightarrow \infty$, we get, for sufficiently small Δ , a discrete-time contract that satisfies all *RDIR* constraints and delivers an expected utility for P arbitrarily close to $J(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**})$, contradicting the fact that P 's optimal discrete-time contract has a utility converging to $J(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$. Therefore, no such $(\tau^{**}, d_{\tau^{**}}^{**}, \alpha_{\tau^{**}}^{**})$ exists, and $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$ must be an optimal contract.

We note $\bar{R}^* < \bar{Y}$; otherwise, if $\bar{R}^* \geq \bar{Y}$, the outside option will never be taken, as $\bar{Y}$ is never reached in finite time, in which case, by the same arguments as in Lemma A.6, all players could be made better-off reaching a split immediately. Finally, the conditions for S^* to be strictly decreasing or constant are shown in Lemma O.A.3 in the online Appendix, while α^* not being constant follows from the fact that $\alpha^*(\bar{R}^*) = 0$ ⁴⁸ and $\alpha^*(m) > 0$ at some $m < \bar{R}^*$, as P can generate a strictly positive expected utility at $m < \bar{R}^*$. ■

REFERENCES

- Altman, Eitan. 1999. *Constrained Markov Decision Processes*. Boca Raton, FL: CRC Press.
- Ausubel, Lawrence M., and Raymond J. Deneckere. 1989. "A Direct Mechanism Characterization of Sequential Bargaining with One-Sided Incomplete Information." *Journal of Economic Theory* 48 (1): 18–46.

⁴⁶Here, we are using the assumption made at the beginning of the Appendix that u_i' and u_i'' are bounded away from zero and ∞ .

⁴⁷Note that in our optimal contract, the restriction to $\alpha_{\tau} \geq 0$ was not binding; even if P could choose $\alpha_{\tau} < 0$, the monotonicity of $\alpha^{\Delta}(m)$ and the fact that $V(m, m) = m$ implies that, whenever $\alpha^{\Delta}(m) < 0$, P would be better-off taking his outside option at (m, m) . Thus, the optimal contract with these extended utility functions is the same as when restricting to $\alpha_{\tau} \in [0, 1]$.

⁴⁸If $\alpha^*(\bar{R}^*) > 0$, then the same argument made to improve upon the stationary contract in Section II implies that an improving continuation contract exists at $\tau_+(\bar{R}^*)$, contradicting the optimality of $(\tau^*, d_{\tau^*}^*, \alpha_{\tau^*}^*)$.

- Backus, Matthew, Thomas Blake, Brad Larsen, and Steven Tadelis. 2020. "Sequential Bargaining in the Field: Evidence from Millions of Online Bargaining Interactions." *Quarterly Journal of Economics* 135 (3): 1319–61.
- Bhattacharya, Rabi N., and Edward C. Waymire. 2009. *Stochastic Processes with Applications*. Philadelphia, PA: Society for Industrial and Applied Mathematics.
- Bird, Daniel, and Alexander Frug. 2022. "Monotone Contracts." *Theoretical Economics* 17 (3): 1041–73.
- Chaves, Isaias N. 2019. "Privacy in Bargaining: The Case of Endogenous Entry." Unpublished.
- Compte, Olivier, and Philippe Jehiel. 2002. "On the Role of Outside Options in Bargaining with Obstinate Parties." *Econometrica* 70 (4): 1477–1517.
- Daley, Brendan, and Brett Green. 2020. "Bargaining and News." *American Economic Review* 110 (2): 428–74.
- DeMarzo, Peter M., and Yuliy Sannikov. 2017. "Learning, Termination, and Payout Policy in Dynamic Incentive Contracts." *Review of Economic Studies* 84 (1): 182–236.
- Dixit, Avinash, Gene M. Grossman, and Faruk Gul. 2000. "The Dynamics of Political Compromise." *Journal of Political Economy* 108 (3): 531–68.
- Durandard, Théo. 2023. "Dynamic delegation in promotion contests." <https://arxiv.org/abs/2308.05668>.
- Ferguson, Merideth, Neta Moye, and Ray Friedman. 2008. "The Lingering Effects of the Recruitment Experience on the Long-Term Employment Relationship." *Negotiation and Conflict Management Research* 1 (3): 246–62.
- Forand, Jean Guillaume, and Jan Zápal. 2020. "Production Priorities in Dynamic Relationships." *Theoretical Economics* 15 (3): 861–89.
- Fuchs, William, and Francesco Lippi. 2006. "Monetary Union with Voluntary Participation." *Review of Economic Studies* 73 (2): 437–57.
- Fuchs, William, and Andrzej Skrzypacz. 2010. "Bargaining with Arrival of New Traders." *American Economic Review* 100 (3): 802–36.
- Fudenberg, Drew, and Luis Rayo. 2019. "Training and Effort Dynamics in Apprenticeship." *American Economic Review* 109 (11): 3780–3812.
- Gryglewicz, Sebastian, and Barney Hartman-Glaser. 2020. "Investment Timing and Incentive Costs." *Review of Financial Studies* 33 (1): 309–57.
- Hu, Zhenyu, and Wenjie Tang. 2021. "Size Matters, So Does Duration: The Interplay Between Offer Size and Offer Deadline." *Management Science* 67 (8): 4935–60.
- Hwang, Ilwoo, and Fei Li. 2017. "Transparency of Outside Options in Bargaining." *Journal of Economic Theory* 167: 116–47.
- Kushner, Harold Joseph, ed. 1977. *Probability Methods for Approximations in Stochastic Control and for Elliptic Equations*. New York, NY: Academic Press.
- Lee, Jihong, and Qingmin Liu. 2013. "Gambling Reputation: Repeated Bargaining with Outside Options." *Econometrica* 81 (4): 1601–72.
- Lippman, Steven A., and John W. Mamer. 2012. "Exploding Offers." *Decision Analysis* 9 (1): 6–21.
- McClellan, Andrew. 2022. "Experimentation and Approval Mechanisms." *Econometrica* 90 (5): 2215–47.
- Merlo, Antonio, and Charles Wilson. 1995. "A Stochastic Model of Sequential Bargaining with Complete Information." *Econometrica* 63 (2): 371–99.
- Nash, John. 1950. "The Bargaining Problem." *Econometrica* 18 (1): 155–62.
- Ortner, Juan. 2017. "Durable Goods Monopoly with Stochastic Costs." *Theoretical Economics* 12 (2): 817–61.
- Peskir, Goran, and Albert Shiryaev. 2006. *Optimal Stopping and Free-Boundary Problems*. Basel: Birkhäuser Verlag.
- Ray, Debraj. 2002. "The Time Structure of Self-Enforcing Agreements." *Econometrica* 70 (2): 547–82.
- Skarlicki, Daniel P., and Robert Folger. 1997. "Retaliation in the Workplace: The Roles of Distributive, Procedural, and Interactional Justice." *Journal of Applied Psychology* 82 (3): 434–43.
- Spear, Stephen E., and Cheng Wang. 2005. "When to Fire a CEO: Optimal Termination in Dynamic Contracts." *Journal of Economic Theory* 120 (2): 239–56.
- Thomas, Jonathan, and Tim Worrall. 1988. "Self-Enforcing Wage Contracts." *Review of Economic Studies* 55 (4): 541–54.
- Zorc, Saša, and Ilija Tsetlin. 2020. "Deadlines, Offer Timing, and the Search for Alternatives." *Operations Research* 68 (3): 927–48.